

INSTITUTE OF ENGINEERING & MANAGEMENT

SALT LAKE, KOLKATA

Affiliated to MAKAUT and approved by AICTE



Syllabus for Autonomous Batch

B.TECH

INFORMATION TECHNOLOGY

INSTITUTE OF ENGINEERING & MANAGEMENT

Semester III (Second year] Curriculum								
Sl. No.	**Type of course**	**Code**	**Course Title**	**Hours per week**				**Credits**
				Lecture	**Tutorial**	**Practical**	**Sessional**	
Theory Papers								
1	Engineering Science Course	ESC301	Analog Electronic Circuits	3	0	0	0	3
2	Professional Core Courses	PCC-CS301	Data structure & Algorithms	3	0	0	0	3
3	Professional Core Courses	ESC 302	Digital Electronics	3	0	0	0	3
4	Professional Core Courses	PCCCS302	IT Workshop	1	0	0	0	1
5	Basic Science course	BSC301	Mathematics - III	3	0	0	0	3
6	Humanities & Social Sciences including Management courses	HSMC 301	Humanities-I (Technical Report Writing)	3	0	0	0	3
7	Humanities and social sciences including Management	HSMC302	Essential Studies for Professionals - III	2	0	0	0	2
	Total			**17**	**0**	**0**	**0**	**18**
Practical Papers								
1	Engineering Science Course	ESC 391	Analog Electronic Circuits Lab	0	0	4	0	2
2	Professional Core Courses	PCC-CS391	Data structure & Algorithms Lab	0	0	4	0	2
3	Professional Core Courses	ESC 392	Digital Electronics Lab	0	0	4	0	2
4	Professional Core Courses	PCC-CS392	IT Workshop Lab	0	0	4	0	2
	Total			**0**	**0**	**16**	**0**	**8**
Sessional Papers								
1	Humanities and social sciences including Management	HSMC382	Skill Development for Professionals - III	0	0	0	2	1
2	Innovative Project	PROJ –CS301	Innovative Project – I	0	0	0	0	1
3	Mandatory Additional Requirements (MAR)	MAR381	Mandatory Additional Requirements (MAR)-III	0	0	0	0	0
4	MOOCs (Mandatory for Honours)	MOOCs 321	Massive Open Online Course 3.1 (Mandatory for B.Tech (Honours))	0	0	0	1	.5
5		MOOCs 322	Massive Open Online Course 3.2 (Mandatory for B.Tech (Honours))	0	0	0	1	.5
	Total			**0**	**0**	**0**	**4**	**3**

	Total	**17**	**0**	**16**	**4**	**29**

2nd Year 3rd Semester Detailed Syllabus

Course Code : ESC301
Course Title : Analog Electronic Circuits
Credit : **3**

Course Outcome	
CO1.	Understand transitor biasing, transistor models and Power supplies.
CO2.	Understand transistor amplifier, frequency response and feedback concepts
CO3.	Understand Operationl Amplifiers and its applications
CO4.	Understanding Multivibrators using timer IC

Module-I: [10]

1. Filters and Regulators: Capacitor filter, π-section filter, ripple factor, series and shunt voltageregulator, percentage regulation, 78xx and 79xx series, concept of SMPS.

2. Transistor Biasing and Stability: Q-point, Self Bias-CE, Compensation techniques, h-model oftransistors. Expression for voltage gain, current gain, input and output impedance, trans-resistance &trans-conductance; Emitter follower circuits, High frequency model of transistors.

Module -II: [10]

1. Transistor Amplifiers: RC coupled amplifier, functions of all components, equivalent circuit, derivationof voltage gain, current gain, input impedance and output impedance, frequency response characteristics,lower and upper half frequencies, bandwidth, and concept of wide band amplifier.

2. Feedback Amplifiers & Oscillators: Feedback concept, negative & positive feedback, voltage/ current,series/shunt feedback, Berkhausen criterion, Colpitts, Hartley's, Phase shift, Wein bridge and crystaloscillators.

Module -III: [10]

1. Operational Amplifier: Ideal OPAMP, Differential Amplifier, Constant current source (current mirroretc.), level shifter, CMRR, Open & Closed loop circuits, importance of

feedback loop (positive &negative), inverting & noninverting amplifiers, voltage follower/buffer circuit.

2. Applications of Operational Amplifiers: adder, integrator & differentiator, comparator. Instrumentation Amplifier, Log & Anti-log amplifiers, Trans-conductance multiplier, PrecisionRectifier, voltage to current and current to voltage converter, free running oscillator.

Module -IV: [8]

Multivibrator – Monostable, Bistable, Astable multivibrators; Monostable and astable operation using555 timer. Schmitt Trigger.

Text Books:

1.Microelectronic Circuits, Sedra & Smith, Oxford University Press.
2.Integrated Electronics, Milman & Halkias, Mc Graw Hill Company.
3.Electronic devices & Circuits, Balbir Kumar & Shail B. Jain, PHI.
4.Op-amps and Linear IC's, R.A. Gayakwad, PHI.

Reference Books:

1.Microelectronic Circuit- Analysis & Design, Rashid, Cenage Learning.
2.Electronic Circuits: Discrete & Integrated, 3rd Edition, Schilling & Belove, Mc Graw Hill Company.
3.Electronic principles, 6th Edition, Malvino, Mc Graw Hill Company.
4.Operational Amplifier & Linear IC's, Bell, Oxford University Press.
5.2000 Solved Problems in Electronics, Jimmie J. Cathey, Mc Graw Hill Inc.
6.Electronic Devices -System & Application, Robert Diffenderfer, Cengage Learning.
7.Op- Amps & Linear Integrated Circuits, Ravi Raj Dudeja & Mohan Dudeja, Umesh Publication

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Course Code : PCC-CS301
Course Title : Data Structure & Algorithm
Credit : **3**

Course Outcome	
CO1.	Interpret and compute asymptotic notations of an algorithm to analyse the consumption of resources (time/space).
CO2.	Exemplify and implement stack, queue and list ADT to manage the memory using static and dynamic allocations. Implement Principles of recursion – use of stack, differences between recursion and iteration
CO3.	Identify appropriate data structure and algorithm for a given contextual problem and develop in C and/or Python. Implement binary search tree to design applications like expression trees. Identify, model, solve and develop code for real life problems like shortest path and MST using graph theory.

CO4.	Develop and compare the comparison-based search algorithms and sorting algorithms. Describe the hash function and concepts of collision and its resolution methods.

Module –I: [8L]

Linear Data Structure

Introduction (2L): Why we need data structure? Concepts of data structures: a) Data and data structure b) Abstract Data Type and Data Type. Algorithms and programs, basic idea of pseudo-code. Algorithm efficiency and analysis, time and space analysis of algorithms – order notations.

Array (2L): Different representations – row major, column major. Sparse matrix - its implementation and usage. Array representation of polynomials.

Linked List (4L): Singly linked list, circular linked list, doubly linked list, linked list representation of polynomial and applications.

Module -II: [7L]

Linear Data Structure

Stack and Queue (5L): Stack and its implementations (using array, using linked list), applications. Queue, circular queue, dequeue. Implementation of queue- both linear and circular (using array, using linked list), applications.

Recursion (2L): Principles of recursion – use of stack, differences between recursion and iteration, tail recursion. Applications - The Tower of Hanoi, Eight Queens Puzzle.

Module –III: [15L]

Nonlinear Data structures

Trees (9L): Basic terminologies, forest, tree representation (using array, using linked list). Binary trees - binary tree traversal (pre-, in-, post- order), threaded binary tree (left, right, full) - non-recursive traversal algorithms using threaded binary tree, expression tree. Binary search tree- operations (creation, insertion, deletion, searching). Height balanced binary tree – AVL tree (insertion, deletion with examples only). B- Trees – operations (insertion, deletion with examples only).

Graphs (6L): Graph definitions and concepts (directed/undirected graph, weighted/un-weighted edges, sub-graph, degree, cut vertex/articulation point, pendant node, clique, complete graph, connected components – strongly connected component, weakly connected component, path, shortest path, isomorphism). Graph representations/storage implementations – adjacency matrix, adjacency list, adjacency multi-list.

Graph traversal and connectivity – Depth-first search (DFS), Breadth-first search (BFS) – concepts of edges used in DFS and BFS (tree-edge, back-edge, cross-edge, forward-edge), applications.

Minimal spanning tree – Prim's algorithm (basic idea of greedy methods).

Module – IV: [10L]

Searching, Sorting

Sorting Algorithms (5L): Bubble sort and its optimizations, insertion sort, shell sort, selection sort, merge sort, quick sort, heap sort (concept of max heap, application – priority queue), radix sort.

Searching (2L): Sequential search, binary search, interpolation search.

Hashing (3L): Hashing functions, collision resolution techniques.

Course Code : ECS302
Course Title : Digital Electronics
Credit : 3

Course Outcome	
CO1.	Students would be able to convert from one number system to another, work out and design problems related to Boolean algebra, minimization etc.
CO2.	Have the ability to identify basic requirements for a design application and propose a cost-effective solution.
CO3.	Have the ability to understand, analyze and design various combinational and sequential circuits.
CO4.	Have the ability to understand, analyze and design various A/D and D/A conversion techniques

Module I:

Binary Number System & Boolean Algebra (recapitulation) [1L]; BCD, ASCII, EBCDIC, Gray codes and their conversions [2L]; Signed binary number representation with 1's and 2's complement methods [1L], Binary arithmetic, Venn diagram, Boolean algebra (recapitulation) [1L]; Representation in SOP and POS forms [1L]; Minimization of logic expressions by KMAP [3L] Quine-McCluskey Minimization Technique (Tabular Method) [2L]

Module II:

Combinational circuits - Adder and Subtractor circuits (half & full adder & subtractor) [2L]; Encoder, Decoder, Comparator, Multiplexer, De-Multiplexer and Parity Generator [4L].

Module III:

Sequential Circuits - Basic Flip-flop & Latch [1L], Flip-flops -SR, JK, D, T and JK Master-slave Flip Flops [3L]Registers (SISO,SIPO,PIPO,PISO) [2L],Ring counter, Johnson counter [1L], Basic concept of Synchronous and Asynchronous counters (detail design of circuits excluded), [2L], Design of Mod N Counter [2L]

Module IV:

A/D and D/A conversion techniques – Basic concepts (D/A :R-2-R only [2L]A/D: successive approximation [2L]) (4L)Logic families- TTL, ECL, MOS and CMOS - basic concepts.(2L)

Course Code : PCCCS302
Course Title : IT Workshop
Credit : 1

Course Outcome	
CO1.	To introduce the basics of scripting & the contributions of scripting languages
CO2.	Design real-life problems and think creatively about solutions
CO3.	Real-life problem solving using R/Matlab/Python programming constructs
CO4.	To be exposed to advanced applications of mathematics, engineering, and natural sciences to program real-life problems

Programming in MATLAB

Module I:

Introduction
Why MATLAB? History, its strengths, Competitors, Starting MATLAB, Using MATLAB as a calculator, Quitting MATLAB

Basics
Familiar with MATLAB windows, Basic Operations, MATLAB-Data types, Rules about variable names, Predefined variables

Module II :

Programming-I
Vector, Matrix, Array Addressing, Built-in functions, Mathematical Operations, Dealing with strings (Array of characters), Array of array (cell) concept

Programming-II
Script file, Input commands, Output commands, Structure of function file, Inline functions, feval command, Comparison between script file and function file

Module III :

Conditional statements and Loop

Relational and Logical Operators, if-else statements, switch-case statements, for loop, while loop, Special commands (break and continue), Import data from a large database, Export data to own file or database

Module IV :

2D Plotting

In-built functions for plotting, Multiple plotting with special graphics, Curve fitting, Interpolation, Basic fitting interface

3D Plotting

Use of meshgrid function, Mesh plot, Surface plot, Plots with special graphics

Text book and Reference books:

1. A Guide to MATLAB: For Beginners and Experienced Users by Brian R. Hunt, Jonathan Rosenberg, and Ronald L Lipsman
2. MATLAB: Easy Way of Learning by S. Swapna Kumar and S. V. B. Lenina
3. MATLAB Programming for Engineers by Stephen J. Chapman
4. MATLAB for Beginners: A Gentle Approach by Peter Issa Kattan

Course Code : BSC301
Course Title : Mathematics & Statistics - III
Credit : 2

Course Outcome	
CO1.	Understanding the concept of Complex analysis and its various techniques
CO2.	Understanding Random Variable & Probability Distributions
CO3.	Comprehensive knowledge of Basic Statistics
CO4.	Understanding Bivariate Distributions

Module I : (Lecture Hours 16)

Complex analysis

Differentiation, Cauchy-Riemann equations, analytic functions, harmonic functions, finding harmonic conjugate; elementary analytic functions (exponential, trigonometric, logarithm) and their properties; Conformal mappings, Mobius transformations and their

properties.
Contour integrals, Cauchy-Goursat theorem (without proof), Cauchy Integral formula (without proof), Liouville's theorem and Maximum-Modulus theorem (without proof); Taylor's series, zeros of analytic functions, singularities, Laurent's series; Residues, Cauchy Residue theorem (without proof), Evaluation of definite integral involving sine and cosine, Evaluation of certain improper integrals using the Bromwich contour.

Module II : (12 lectures)

Random Variable & Probability Distributions

Discrete random variables, Independent random variables, the multinomial distribution, Poisson approximation to the binomial distribution, infinite sequences of Bernoulli trials, sums of independent random variables; Expectation of Discrete Random Variables, Moments, Variance of a sum, Correlation coefficient, Chebyshev's Inequality. Continuous random variables and their properties, distribution functions and densities, normal, exponential and gamma densities.

Module III : (8 lectures)

Basic Statistics:

Measures of Central tendency: Moments, skewness and Kurtosis - Probability distributions: Binomial, Poisson and Normal - evaluation of statistical parameters for these three distributions, Correlation and regression – Rank correlation.

Module IV : (lecture Hours 4)

Bivariate Distributions:

Bivariate distributions and their properties, distribution of sums and quotients, conditional densities, Bayes' rule.

Suggested Text/Reference Books

1. G.B. Thomas and R.L. Finney, Calculus and Analytic geometry, 9th Edition, Pearson, Reprint,2002.
2. Erwin kreyszig, Advanced Engineering Mathematics, 9th Edition, John Wiley & Sons, 2006.
3. J. W. Brown and R. V. Churchill, Complex Variables and Applications, 7th Ed., Mc-Graw Hill,2004.
4. N.P. Bali and Manish Goyal, A text book of Engineering Mathematics, LaxmiPublications, Reprint, 2008.
5. B.S. Grewal, Higher Engineering Mathematics, Khanna Publishers, 36th Edition, 2010.
6. P. G. Hoel, S. C. Port and C. J. Stone, Introduction to Probability Theory, Universal Book Stall, 2003(Reprint).

7. S. Ross, A First Course in Probability, 6th Ed., Pearson Education India, 2002.
8. W. Feller, An Introduction to Probability Theory and its Applications, Vol. 1, 3rd Ed., Wiley, 1968.
9. Lipschutz S., and Lipson M.L.: Probability (Schaum's Outline Series), TMH.
10. John E. Freund, Ronald E. Walpole, Mathematical Statistics, Prentice Hall

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Course Code : HSMC 301
Course Title : Humanities – 1
Credit : 3

Course Outcome	
CO1.	Ability to recognize, explain, and use the rhetorical strategies and the formal elements of these specific genres of technical communication: technical abstracts, database research reports, instructional manuals, technical descriptions, web pages, wikis, and correspondence.
CO2.	Ability to use the unique qualities of professional reading, speaking style, such as sentence conciseness, clarity, accuracy, honesty, avoiding wordiness or ambiguity, using direct order organization, readability, coherence and transitional devices; ability to revise and edit effectively in all assignments.
CO3.	Develop professional work habits, including those necessary for effective collaboration and cooperation with other students, instructors and professionals.
CO4.	Apply technical information and knowledge in practical presentation for a variety of professional audiences (including peers and colleagues or management) and for public audiences.

Module I - CO1

A. Technical Report Writing -

1. Report Types (Organizational / Commercial / Business / Project)

2. Report Format & Organization of Writing Materials

3. Report Writing (Practice Sessions & Workshops using LaTex)

Module II – CO2

B. Language Laboratory Practice -

Introductory Lecture to help the students get a clear idea of Technical Communication & the need of Language Laboratory Practice Sessions

Conversation Practice Sessions: (To be done as real-life interactions)

a) Training the students by using Language Lab Device/Recommended Texts/cassettes /cd's to get their Listening Skill & Speaking Skill honed

b) Introducing Role Play & honing over all Communicative Competence

Module III – CO3

Group Discussion Sessions:

a) Teaching Strategies of Group Discussion

b) Introducing Different Models & Topics of Group Discussion

c) Exploring Live /Recorded GD Sessions for mending students' attitude/approach & for taking remedial measure

Interview Sessions;

a) Training students to face Job Interviews confidently and successfully

b) Arranging Mock Interviews and Practice Sessions for integrating Listening Skill with Speaking Skill in a formal situation for effective communication.

Module IV – CO4

Presentation-

a) Teaching Presentation as a skill

b) Strategies and Standard Practices of Individual /Group Presentation

c) Media & Means of Presentation: OHP/POWER POINT/ Other Audio-Visual Aids

Competitive Examination-

a) Making the students aware of Provincial /National/International Competitive Examinations

b) Strategies/Tactics for success in Competitive Examinations

c) SWOT Analysis and its Application in fixing Target

Learning Resources:

1. Adrian Duff et. al. (ed.): Cambridge Skills for Fluency –

A) Speaking (Levels 1-4 Audio Cassettes/Handbooks)

B) Listening (Levels 1-4 Audio Cassettes/Handbooks) Cambridge University Press 1998

2. Mark Hancock: English Pronunciation in Use 4 Audio Cassettes/CD'S OUP 2004

Course Code : HSMC302
Course Title : Essential Studies For Professionals - III

Credit : 2
Pre-Requisites : Basic Social Science from primary to high school, NCERTs
L-T-P : 2-0-0 (Total Contact Hrs. 2)

Course Objective:

1. To learn about basic of History to know about our past and to implement it in our daily life.
2. To learn about the Political System of Our Country.
3. To learn the concepts of Basics of Geography and Economics from which Students will acquire knowledge for Competitive exams.

Course Outcome	
At the end of the course the students will be able	
CO1.	To inculcate human values and ethical thinking among students.
CO2.	To prepare the stage for facing different levels of civil service and other competitive examinations.
CO3.	To prepare the ground for making them aware of the happenings, cultural historical and developmental aspects of the country as well as global affairs
CO4.	Learning current affairs with technique.

Course Content:

Module No.	Description	Hours	Blooms Level	PO(1..12) Mapping
CO1.	**Laws of Society** : Union Executive- President, Vice President, PM and Council of Ministers, Attorney General	6	L1 (Remember) L2 (Understand) L4 (Analyse)	PO6, PO7, PO8
CO2.	**Our Freedom Struggle**: Arrival of the Europeans- Portuguese, Dutch, English, French; Land Revenue System, Economic Exploitation of British Rule, Socio-religious Reforms Movement.	12	L1 (Remember) L2 (Understand) L3 (Apply) L4 (Analyze)	PO6, PO7, PO8
CO3.	**Know Our Country:** Physical Geography of India- Peninsular Plateau, Northern Great Plains, Coastal Plains, Soil of India.	12	L1 (Remember) L2 (Understand) L3 (Apply) L4 (Analyze)	PO6, PO7, PO8

CO4.	**RBI and Banking, India and World and Universal Human Values:** Banking System of India with reference to RBI, Capital Market Monthly Current Affairs Magazine and Understanding Human Beings as the co-existence of the self and the body, Program to ensure self-regulation and health, Understanding harmony in the nature.	6	L1 (Remember) L2 (Understand) L3 (Apply) L4 (Analyze)	PO6, PO7, PO8

Learning Resources:

Text Books:

1. NCERT Books from class 8-12.

Reference Books:

1. Indian Constitution- M.Laxmikant
2. Indian Economy-Ramesh Singh
3. History of Modern India- Bepan Chandra
4. Geography of India- Majid Hussain
5. Current Affairs Magazine of IEM-UEM

Course Code : ESC391
Course Title : Analog Electronic Circuits Laboratory
Credit : 2

Course Outcome	
CO1.	Identify relevant information to supplement to the Analog Electronic Circuit ESC 301 course.
CO2.	Set up testing strategies and experiment procedures, then select proper instruments to evaluate performance characteristics of electronic circuit and analyze their operation different operating conditions.
CO3.	Evaluate possible causes of discrepancy in practical experimental observations in comparison to theory.
CO4.	Prepare professional quality textual and graphical presentations of laboratory data and Computational results, incorporating accepted data analysis and synthesis methods, Mathematical software and word-processing tools.

List of Experiment:

1. Study of Clipper and Clamper circuit.
2. Study of Ripple and Regulation characteristics of full wave rectifier with and without capacitor filter.
3. Study of Zener diode as voltage regulator.

4. Construction of two stages R-C coupled amplifier & study of its gain and Bandwidth.
5. Realization V-I & I-V converter using Operational Amplifier.
6. Study of timer circuit using NE 555 and configuration of Monostable and Astable Multivibrator.
7. Study of DAC & ADC
8. Study of differentiator and integrator circuit using Op-Amp
9. Study of Inverting and Non Inverting Amplifier using Op-Amp
10. Study of Voltage Adder-Difference Circuit using Op-Amp
11. Study of comparator and Schmitt trigger using Op-Amp
12. Study of Phase shift oscillator using OPAMP

Course Code : PCC CS391
Course Title : Data Structure & Algorithm Lab
Credit : 2

Experiments should include but not limited to:

Course Outcome

CO1.	Familiarity to map real world problems into programming language. Implement basic data structures such as arrays and linked list.
CO2.	Can solve the problems in systematic way. Applications of linked list like polynomial addition and multiplication, addition and multiplication of sparse matrices, Josephus problem.
CO3.	Efficiently implement linear, non-linear data structures .Programs to demonstrate the implementation of various operations on stack and queue.
CO4.	Implement various Binary trees, Binary search trees, searching and sorting algorithms. Programs to demonstrate fundamental algorithmic problems including Tree Traversals, Graph traversals and shortest paths.

Data Structure & Algorithm Lab SYLLABUS:

Module I:

Implementation of array operations, iterative and recursive functions.
Implementation of different operations on Single, Double and Circular Linked List like insertion, deletion, inversion, merging, traversal.

Module II:

Implementation of applications of linked list like polynomial addition and multiplication, addition and multiplication of sparse matrices, Josephus problem. Evaluation of expressions operations on Multiple stacks & queues.

Module III:

Introduce and Implementation the concept of data structures through ADT including Stacks, Queue, Circular Queue using arrays and linked list.

Module IV:

Application of Trees, Recursive and No recursive traversal of Trees, AVL Tree. Application of sorting and searching algorithms.
Hash tables implementation: searching, inserting and deleting, searching & sorting techniques.

Course Code : ECS392
Course Title : Digital Electronics Laboratory
Credit : 2

Course Outcome	
CO1.	Ability to design and implement both combinational and sequential circuits and to analyze their operations.
CO2.	Ability to solve engineering problems in digital system design.
CO3.	Have the ability to identify basic requirements for a design application and propose a cost-effective solution.
CO4.	Communicate effectively about laboratory work both orally and in writing journals/technical reports.

SYLLABUS:

1) Familiarity with basic gates ICs and Realization of NOT, AND, OR and XOR operations by using universal gates (both NAND and NOR). - Design some basic logic circuits using basic gates ICs.
2) Design a circuit to indicate 4 bit odd and even numbers.
3) Realization of a circuit to display prime and non prime numbers (4 bit).
4) Implementation of Half Adder. Implementation of Full Adder. Carryout expression is implemented by basic gates.
5) Implementation of Full Adder by using 2 half Adders and an OR gate.
6) Implementation of Half Subtractor.
7) Implementation of Full Subtractor. Borrow out expression is implemented by basic gates.
8) Implementation of Full Subtractor using 2 Half Subtractors and an OR gate.
9) Realization of a circuit to convert BCD to Excess -3 codes.
10) Realization of a circuit to convert Excess -3 codes to BCD.
11) Design a circuit to convert 4 bit Binary to 4 bit Gray code.
12) Design a circuit to convert 4 bit Gray code to 4 bit Binary.
13) Realization of an Even Parity Generator and Checker circuit.
14) Implementation of 2 bit comparator circuit.
15) Realization of the internal architecture of 4:1 Multiplexer and 1:4 De-multiplexer.

16) Implementation of Full Adder using MUX IC 4539B.
17) Implementation of Full Subtractor using MUX IC 4539B.
18) Realization of 4:2 Priority Encoder along with output indicator (basic gates).
19) Realization of the internal architecture of 3:8 Decoder using basic gates.
20) Realization of octal to binary encoder using basic gates.
21) Implement Full Adder using IC 4008.
22) Implement Full Subtractor using IC 4008.
23) Truth table verification of SR flip-flop (using NAND gates only).
24) Truth table verification of D flip-flop (using NAND gates only).
25) Truth table verification of JK flip-flop (using NAND gates only).
26) Truth table verification of T flip-flop (using NAND gates only).
27) Design a Master slave flip-flop.
28) Design 4-bit synchronous up counter.
29) Design 4-bit synchronous down counter.
30) Design 4-bit asynchronous up counter.
31) Design 4-bit asynchronous down counter.
32) Design a 3-bit synchronous up/down' counter using JK flip-flop with external mode signal M. If M=1,counter counts up and with M=0, counter counts down.
33) Design and implement MOD-4 Ring counter.
34) Design a Johnshon counter.
35) Design a Decade counter.
36) Design an unit distance code counter.
37) Realization of Serial-in-Serial-Out shift register.
38) Realization of Serial-In-Parallel Out Shift register.
39) Realization of Parallel-In-Parallel Out Shift register.
40) Realization of Parallel-In-Serial Out Shift register.
41) Realization of Bidirectional shift register (All using D flip-flops).

Course Code : PCC-CS392
Course Title : IT Workshop – (Scilab/MATLAB) Lab
Credit : 2

Course Outcome

CO1.	To make the students aware about the Scilab software environment.
CO2.	To make the students able to implement the basics of Scilab programming
CO3.	To make the students able to apply the knowledge of Scilab programming for different engineering applications
CO4.	To enable the students to learn and apply Scilab Simulink for simulation, analysis and design of the system

SYLLABUS:

Module	Description	Lec-Hr
1	Introduction to Scilab and its benefits. Installing Scilab. Introduction to Toolboxes.	2

2	Basic functions. Vector Operations, length of a vector, mathematical operations on Vectors such as addition, subtraction and multiplication. Matrix Operations, access the elements of Matrix, determine the determinant, inverse and eigen values of a matrix, special matrices, elementary row operation, Conditional Branching-if,then,else,elseif keyword, for statement, Scripts and Functions, introduction to the file formats in Scilab. script files, sce and. sci , inline functions, Plotting 2D graphs. File Handling- Scilab. User Defined Input and Output in Scilab Input Function.	10
3	Develop Scilab code for different Composite Numerical Integration algorithms. Apply the algorithm to each interval. Numerical methods- Solving Non- linear Equations Solve nonlinear equations using numerical methods, Bisection method, Secant method. Gauss Elimination method algorithm, Gauss Jordan method algorithm, linear equations using iterative methods, Jacobi and Gauss Seidel iterative methods. Develop Scilab code for different Numerical Interpolation algorithms, ODEs using Euler Methods, ODEs using Euler and Modified Euler methods, Xcos, Palette.	10
4	Use of Scilab function Karmarkar in Optimization, Plotting continuous and discrete sine waves. Plotting step function. Plotting ramp function, continuous time system: second and higher order, Response plot for step input, sine input and bode plot, discrete time system variable z, first order discrete time system, flts, dscr, ss2tf functions, Use of scifunc block in XCOS , Use of MUX block , Simulating a PID controller using Xcos, Scilab Toolbox for calling external C libraries, Scilab Toolbox for calling Python and its functions	10

Course Code : HSMC382
Course Title : **SKILL DEVELOPMENT FOR PROFESSIONALS - III**
Credit : 1
Pre-Requisites : Basic Mathematics, General English from primary to high school.
L-T-P : 2-0-0 (Total Contact Hrs. 2)

Course Objective:

1. To enhance the aptitude & analytical skill of students with multiple tricky approaches.
2. To prepare the students for various competitive examinations & professional exams

Course Outcome

At the end of the course the students will be able

CO1.	To enhance their problem solving skills, to improve the basic mathematical & Logical Skills for any type of competitive examinations.
CO2.	To get best possible training for the them students through continuous training module.
CO3.	To find themselves sound for the campus recruitment program's aptitude Test.

CO4.	To enhance problem solving skill using fast track techniques without using calculator.

Course Content:

Module No.	Description	Hours	Blooms Level	PO(1..12) Mapping
1.	**Quantitative Aptitude** Simple & Compound Interest, Data Interpretation, Indices & Surds, Number System, Quadratic Equations	6	L1 (Remember) L2 (Understand) L4 (Analyse)	PO1, PO2, PO10
2.	**Logical Reasoning: Syllogism,** Logical Venn diagram, If Else Statement **Puzzles** Seating Arrangement, Classification, Seating Arrangement with Blood Relations **Machine Input-Output** Pattern Based I/O **Inequality** a) Coded Inequality, b) Jumbled Inequality, c) Conditional inequality.	12	L1 (Remember) L2 (Understand) L3 (Apply) L4 (Analyze)	PO1, PO2, PO10
3.	**Verbal English** **Sentence Corrections,** Fill the **blanks** with appropriate words/ articles/ preposition/verbs/adverbs/conjunction**.** **Reading Comprehension** (Advance Level) **Vocabulary**.	12	L1 (Remember) L2 (Understand) L3 (Apply) L4 (Analyze)	PO1, PO2, PO10
4.	**Data interpretation:** Advanced Level.	6	L1 (Remember) L2 (Understand) L3 (Apply) L4 (Analyze)	PO1, PO2, PO10

Learning Resources:

Reference Books:

1. Objective General English- S.P Bakshi
2. English Grammar and Competition-S.C Gupta
3. Fast Track Objective Arithmetic- Rajesh Verma
4. Advance Maths- Rakesh Yadav
5. Verbal and Non-Verbal Reasoning- R.S Agarwal
6. A new approach to Reasoning- BS Sijwali
7. Quantitative Aptitude-R.S Agarwal

Semester IV (Second year) Curriculum								
Sl. No.	**Type of course**	**Code**	**Course Title**	**Hours per week**				**Credits**
				Lecture	**Tutorial**	**Practical**	**sessional**	
Theory Papers								
1	Professional Core Courses	PCCCS401	Discrete Mathematics	3	0	0	1	4
2	Engineering Science Courses	PCCCS402	Computer Organization & Architectur	3	0	0	0	3
3	Professional Core Courses	PCCCS403	Operating Systems	3	0	0	0	3
4	Professional Core Courses	PCCCS404	Design & Analysis of Algorithms	3	0	0	0	3
5	Professional Core Courses	PCCCS405	Artificial Intelligence & Machine Learning	2	0	0	0	2
6	Basic Science Course	BSC401	Mathematics IV	3	0	0	0	3
7	Humanities & Social Sciences including Management Courses	HSMC401	Management 1 (Organizational Behaviour)	3	0	0	0	3
8	Humanities & Social Sciences including Management courses	HSMC402	Essential Studies for Professionals - IV	2	0	0	0	0.5
8	Mandatory Courses	MC401	Environmental Sciences	**2**	**0**	**0**	**0**	**0**
	Total			24	0	0	1	21.5
Practical Papers								
9	Engineering Science Courses	PCCCS492	Computer Organization & Architectur	0	0	4	0	2
10	Professional Core Courses	PCCCS493	Operating Systems	0	0	4	0	2
11	Professional Core Courses	PCCCS494	Design & Analysis of Algorithms	0	0	4	0	2
12	Professional Core Courses	PCCCS495	Artificial Intelligence & Machine Learning	0	0	2	0	1
Total				0	0	14	0	7
Sessional Papers								
13	Humanities & Social Sciences including	HSMC482	Skill Development for Professionals -	0	0	0	2	0.5

	Management courses		IV					
14	Innovative Project	PROJCS401	Innovative Project – II (Problem Solving Approaches using Design Patterns)	0	0	0	0	1
15	Mandatory Additional Requirements(MAR)	MAR481	Mandatory Additional Requirements (MAR)-IV	0	0	0	0	0
Total				0	0	0	2	1.5
Total								**30**

Twenty Credit Points are required in Massive Open Online Courses (MOOCs).

Twenty certificates are required in International Foreign Certification.

Course Name: Discrete Mathematics

Course Code: PCCCS401

Semester: 4

Contact Hours: L-T-P: 3-1-0

Credits: 4

Prerequisite: Linear algebra, Calculus

Module 1 [10L]

Sets, Relation and Function: Operations and Laws of Sets, Cartesian Products, Binary Relation, Partial Ordering Relation, Equivalence Relation, Image of a Set, Sum and Product of Functions, Bijective functions, Inverse and Composite Function, Size of a Set, Finite and infinite Sets, Countable and uncountable Sets, Cantor's diagonal argument and The Power Set theorem, Schroeder-Bernstein theorem.

Principles of Mathematical Induction: The Well-Ordering Principle, Recursive definition, The Division algorithm: Prime Numbers, The Greatest Common Divisor: Euclidean Algorithm, The Fundamental Theorem of Arithmetic.

Module 2 [8L]

Basic counting techniques-inclusion and exclusion, pigeon-hole principle, permutation and combination.

Propositional Logic: Syntax, Semantics, Validity and Satisfiability, Basic Connectives and Truth Tables, Logical Equivalence: The Laws of Logic, Logical Implication, Rules of Inference, The use of Quantifiers. **Proof Techniques:** Some Terminology, Proof Methods and Strategies, Forward Proof, Proof by Contradiction, Proof by Contraposition, Proof of Necessity and Sufficiency.

Module 3 [12L]

Algebraic Structures and Morphism: Algebraic Structures with one Binary Operation, Semi Groups, Monoids, Groups, Congruence Relation and Quotient Structures, Free and Cyclic Monoids and Groups, Permutation Groups, Substructures, Normal Subgroups, Algebraic Structures with two Binary Operation, Rings, Integral Domain and Fields. Boolean Algebra and Boolean Ring, Identities of Boolean Algebra, Duality, Representation of Boolean Function, Disjunctive and Conjunctive Normal Form.

Module 4 [12L]

Graphs and Trees: Graphs and their properties, Degree, Connectivity, Path, Cycle, Sub Graph, Isomorphism, Eulerian and Hamiltonian Walks, Graph Colouring, Colouring maps and Planar Graphs, Colouring Vertices, Colouring Edges, List Colouring, Perfect Graph, definition properties and Example, rooted trees, trees and sorting, weighted trees and prefix codes, Bi-connected component and Articulation Points, Shortest distances.

Text Books and/or Reference Material

Text Books:

1. Kenneth H. Rosen, Discrete Mathematics and its Applications, Tata McGraw – Hill

2. Susanna S. Epp, Discrete Mathematics with Applications,4th edition, Wadsworth Publishing Co. Inc.

3. C L Liu and D P Mohapatra, Elements of Discrete Mathematics A Computer Oriented Approach, 3rd Edition by, Tata McGraw – Hill.

Reference books:

1. J.P. Tremblay and R. Manohar, Discrete Mathematical Structure and Its Application to Computer Science", TMG Edition, TataMcgraw-Hill

2. Norman L. Biggs, Discrete Mathematics, 2nd Edition, Oxford University Press. Schaum's Outlines Series, Seymour Lipschutz, Marc Lipson

3. Discrete Mathematics, Tata McGraw – Hill

Course Name: Computer Organization & Architecture

Course Code: PCCCS402

Semester: 4

Contact Hours: L-T-P: 3-0-0

Credits: 3

Prerequisite: Digital Electronics

Module 1 [12L]

Functional blocks of a computer: CPU, memory, input-output subsystems, control unit. Instruction set architecture of a CPU–registers, instruction execution cycle, RTL interpretation of instructions, addressing modes, instruction set. Case study – instruction sets of some common CPUs.

Data representation: signed number representation, fixed and floating point representations, character representation. Computer arithmetic – integer addition and subtraction, ripple carry adder, carry look ahead adder, etc. multiplication – shift-andadd, Booth multiplier, carry save multiplier, etc. Division restoring and non-restoring techniques, floating point arithmetic.

Module 2 [7L]

Introduction to x86 architecture. CPU control unit design: hardwired and micro-programmed design approaches, Case study – design of a simple hypothetical CPU.

Memory system design: semiconductor memory technologies, memory organization. Peripheral devices and their characteristics: Input-output subsystems, I/O device interface, I/O transfers–program controlled, interrupt driven and DMA, privileged and non-privileged instructions, software interrupts and exceptions. Programs and processes–role of interrupts in process state transitions, I/O device interfaces – SCII, USB

Module 3 [5L]

Pipelining: Basic concepts of pipelining, throughput and speedup, pipeline hazards. Parallel Processors: Introduction to parallel processors, Concurrent access to memory and cache coherency.

Module 4 [6L]

Memory organization: Memory interleaving, concept of hierarchical memory organization, cache memory, cache size vs. block size, mapping functions, replacement algorithms, write policies.

Text Books

1. “Computer Organization and Design: The Hardware/Software Interface”, 5th Edition by David A. Patterson and John L. Hennessy, Elsevier.
2. “Computer Organization and Embedded Systems”, 6th Edition by CarlHamacher, McGraw Hill Higher Education.

Reference Books

1. “Computer Architecture and Organization”, 3rd Edition by John P. Hayes, WCB/McGraw-Hill
2. “Computer Organization and Architecture: Designing for Performance”, 10th Edition by William Stallings, Pearson Education.
3. “Computer System Design and Architecture”, 2nd Edition by Vincent P. Heuring and Harry F. Jordan, Pearson Education.

Course Name: Operating Systems

Course Code: PCCCS403

Semester: 4

Contact Hours: L-T-P: 3-0-0

Credits: 3
Prerequisite: Data Structure & Algorithms

Module 1 [8L]

Introduction: Generations & Concept of Operating Systems, Types of Operating Systems, OS Services, System Calls, Structure of an OS - Layered, Monolithic, Microkernel Operating Systems, Concept of Virtual Machine. Case study on UNIX and WINDOWS Operating System. **Processes:** Definition, Process Relationship, Different states of a Process, Process State transitions, Process Control Block (PCB), Context switchin. **Thread:** Definition, Various states, Benefits of threads, Types of threads, Concept of

multithreads. **Process Scheduling:** Foundation and Scheduling objectives, Types of Schedulers, Scheduling criteria: CPU utilization, Throughput, Turnaround Time, Waiting Time, Response Time; Scheduling algorithms: Pre-emptive and Non pre-emptive, FCFS, SJF, RR; Multiprocessor scheduling: Real Time scheduling: RM and EDF.

Module 2 [10L]

Inter-process Communication: Critical Section, Race Conditions, Mutual Exclusion, Hardware Solution, Strict Alternation, Peterson's Solution, The Producer Consumer Problem, Semaphores, Event Counters, Monitors, Message Passing, Classical IPC Problems: Reader's & Writer Problem, Dinning Philosopher Problem etc. **Deadlocks:** Definition, Necessary and sufficient conditions for Deadlock, Deadlock Prevention, Deadlock Avoidance: Banker's algorithm, Deadlock detection and Recovery.

Module 3 [8L]

Memory Management: Basic concept, Logical and Physical address map, Memory allocation:
Contiguous Memory allocation– Fixed and variable partition– Internal and External fragmentation and Compaction; Paging: Principle of operation –Page allocation Hardware support for paging, Protection and sharing, Disadvantages of paging. **Virtual Memory:** Basics of Virtual Memory – Hardware and control structures – Locality of reference, Page fault, Working Set, Dirty page/Dirty bit – Demand paging, Page Replacement algorithms: Optimal, First in First Out (FIFO), Second Chance (SC), Not Recently used (NRU) and Least Recently used (LRU)

Module 4 [6L]

I/O Hardware: I/O devices, Device controllers, Direct memory access Principles of I/O Software: Goals of Interrupt handlers, Device drivers, Device independent I/O software, Secondary-Storage Structure: Disk structure, Disk scheduling algorithms **File Management:** Concept of File, Access methods, File types, File operation, Directory structure, File System structure, Allocation methods (contiguous, linked, indexed), Free space management (bit vector, linked list, grouping), directory implementation (linear list, hash table), efficiency and performance.
Disk Management: Disk structure, Disk scheduling: FCFS, SSTF, SCAN, C SCAN, Disk reliability, Disk formatting, Boot-block, Bad blocks

Text Books

1. Operating System Concepts Essentials, 9th Edition byAviSilberschatz, Peter Galvin, Greg Gagne, Wiley Asia Student Edition.
2. Operating Systems: Internals and Design Principles, 5th Edition,William Stallings, Prentice Hall of India.

Reference Books

1. Operating System Concepts, Ekta Walia, Khanna Publishing House (AICTE Recommended Textbook – 2018)
2. Operating System: A Design-oriented Approach, 1st Edition by Charles Crowley, Irwin

Publishing
3. Operating Systems: A Modern Perspective, 2nd Edition by Gary J. Nutt, Addison- Wesley
4. Design of the Unix Operating Systems, 8th Edition by Maurice Bach, Prentice-Hall of India
5. Understanding the Linux Kernel, 3rd Edition, Daniel P. Bovet,Marco Cesati, O'Reilly and Associates

Course Name: Design & Analysis of Algorithms

Course Code: PCCCS404

Semester: 4

Contact Hours: L-T-P: 3-0-0

Credits: 3

Prerequisite: Data Structure & Algorithms

Module 1[8L]

Introduction: Characteristics of algorithm. Analysis of algorithm: Asymptotic analysis of complexity bounds – best, average and worst-case behavior; Performance measurements of Algorithm, Time and space trade-offs, Analysis of recursive algorithms through recurrence relations: Substitution method, Recursion tree method and Masters' theorem.

Module 2 [14L]

Fundamental Algorithmic Strategies: Brute-Force, Greedy, Dynamic Programming, Branch-and-Bound and Backtracking methodologies for the design of algorithms;

Illustrations of these techniques for Problem-Solving , Bin Packing, Knap Sack TSP. Heuristics – characteristics and their application domains.

Module 3 [8L]

Graph and Tree Algorithms: Traversal algorithms: Depth First Search (DFS) and Breadth First Search (BFS); Shortest path algorithms, Transitive closure, Minimum Spanning Tree, Topological sorting,Network Flow Algorithm.

Module 4 [6L]

Tractable and Intractable Problems: Computability of Algorithms, Computability classes – P, NP, NP-complete and NP-hard. Cook's theorem, Standard NP-complete problems and Reduction techniques. Advanced Topics: Approximation algorithms, Randomized algorithms, Class of problems beyond NP – PSPACE.

Text Books

1. Introduction to Algorithms, 4TH Edition, Thomas H Cormen, Charles E Lieserson, Ronald L Rivest and Clifford Stein, MIT Press/McGraw-Hill.
2. Fundamentals of Algorithms – E. Horowitz et al.

Reference Material

1. Algorithm Design, 1ST Edition, Jon Kleinberg and ÉvaTardos, Pearson.
2. Algorithm Design: Foundations, Analysis, and Internet Examples, Second Edition, Michael T Goodrich and Roberto Tamassia, Wiley.
3. Algorithms—A Creative Approach, 3RD Edition, UdiManber, Addison-Wesley

Course Name: Artificial Intelligence & Machine Learning

Course Code: PCCCS405

Semester: 4

Contact Hours: L-T-P: 3-0-0

Credits: 2

Prerequisite: Data Structure & Algorithms

Module 1 [5L]

Introduction to Artificial Intelligence: The Foundations of Artificial Intelligence, The History of Artificial Intelligence, and the State of the Art. Knowledge Representation: A Knowledge-Based Agent, Knowledge Representation, Reasoning & Logic, Propositional Logic, Inference in First Order Logic

Module 2 [6L]
Search techniques: AI-Problem formulation, solving problems by searching, uninformed search strategies: depth first search, breadth first search, depth limited search, iterative deepening search, bi-directional search and others.

Module 3 [6L]
Introduction to Machine Learning: Machine learning and it's types; Applications of machine learning; Issues in machine learning. Modelling and Evaluation: Selecting a model; Training model - Holdout, k- fold cross-validation, bootstrap sampling; Model representation and interpretability - under-fitting, over- fitting, bias-variance tradeoff; Model performance evaluation - Classification, regression, clustering; Performance improvement.

Module 4 [5L]
Supervised learning – Classification: k-Nearest Neighbour; Decision tree; Naïve Bayes. Supervised learning – Regression: Linear regression. Unsupervised learning: k-Means, Association Analysis

Text & Reference Books
1. E. Rich and K. Knight, "Artificial intelligence", TMH, 2nd ed., 1992.
2. N.J. Nilsson, "Principles of AI", Narosa Publ. House, 1990.
3. D.W. Patterson, "Introduction to AI and Expert Systems", PHI, 1992.
4. Peter Jackson, "Introduction to Expert Systems", AWP, M.A., 1992.
5. R.J. Schalkoff, "Artificial Intelligence - an Engineering Approach", McGraw Hill Int. Ed., Singapore, 1992.
6. M. Sasikumar, S. Ramani, "Rule Based Expert Systems", Narosa Publishing House, 1994.
7. Machine learning Tom Mitchell, McGraw Hill,1997

Course Name: Mathematics IV

Course Code: BSC 401

Semester: 4

Contact Hours: L-T-P: 3-0-0

Credits: 3

Prerequisite: BSC 103, BSC 104, BSC 301

Module 1: Method of Least Squares and Curve Fitting (6L)

Principle of Least Squares, Curve fitting by the method of least squares- fitting of straight lines, second degree parabolas and exponential curves.

Module 2: Estimation of Parameters (6L)

Point and Interval estimations, Biased and Unbiased estimators, Minimum Variance Unbiased Estimator (MVUE), Consistent Estimator, Maximum Likelihood Estimation of Parameters, Applications in populations following theoretical distributions (Binomial, Poisson and Normal), Calculation of confidence limits for population mean and population proportions.

Module 3: Testing of Hypothesis (Large and Small Sample Tests) (14L)

Large Sample Test: Statistical Hypothesis, Test Statistic, Best Critical Region, Test for single mean, difference of means, single proportion, difference of proportions, and difference of standard deviations. Small Sample Test: Test for single mean, difference of means and correlation coefficients, Test for ratio of variances, Chi-square test for goodness of fit and independence of attributes. One way ANOVA.

Module 4: Transform Calculus (14L)

Laplace Transform: Laplace Transform and its properties, Laplace Transform of periodic functions. Finding Inverse Laplace Transform by different methods, Convolution theorem. Evaluation of integrals by Laplace Transform, Solving ODEs by Laplace Transform method. Fourier Transforms: Fourier Transform and its properties, Fourier Sine and Cosine Transforms, Inverse Fourier Transform (statement only), Fourier Transform of Derivatives (statement only), Convolution theorem (statement only), Related problems and their applications to engineering.

Recommended Text/Reference Books:

1. Sheldon M. Ross (2021), Introduction to Probability and Statistics for Engineers and Scientists, 6th Edition, Academic Press, ISBN: 978-0128243466
2. Douglas C. Montgomery (2017), Applied Statistics and Probability for Engineers, 7th Edition, John Wiley & Sons, ISBN: 978-1119409533
3. Richard A Johnson, Irwin Miller and John Freund (2015), Probability and Statistics for Engineers, 8th Edition, Pearson Education India, ISBN: 978-9332550414
4. Murray R. Spiegel, John Schiller and R. Alu Srinivasan, R. A., (2012), Probability & Statistics- Schaum's Outlines, 4th Edition, McGraw Hill, ISBN: 978-0071795579
5. Ronald E. Walpole, Raymomd H. Myers, Sharon L. Myers and Keying. Ye (2010), Probability and Statistics for Engineers and Scientists, 9th Edition, Pearson Education, ISBN: 978 0321629111
6. S. C. Gupta and V. K. Kapoor (2014), Fundamentals of Mathematical Statistics, 12th Edition, Sultan Chand & Sons, ISBN: 978-8180545283
7. Saktipada Nanda and Sibashis Nanda (2022), A Course on Probability & Statistics, 1st Edition,
Mindprobooks, ISBN: 978-1922608314

8. B. S. Grewal (2021), Higher Engineering Mathematics, 44th Edition, Khanna Publishers, ISBN: 978- 8193328491
9. R. K. Jain and S. R. K. Iyengar (2017), Advanced Engineering Mathematics, 5th Edition, Narosa Publishing House, ISBN: 978-8184875607

Course Name: Management 1 (Organizational Behaviour)

Course Code: HSMC401

Semester: 4

Contact Hours: L-T-P: 3-1-0

Credits: 3

Module 1 [6L]

Organizational Behaviour: Definition, Importance, Historical Background, Fundamental Concepts of OB, Challenges and Opportunities for OB. Personality and Attitudes: Meaning of personality, Personality Determinants and Traits, Development of Personality, Types of Attitudes, Job Satisfaction

Module 2 [8L]

Perception: Definition, Nature and Importance, Factors influencing Perception, Perceptual Selectivity, Link between Perception and Decision Making. Motivation: Definition, Theories of Motivation - Maslow's Hierarchy of Needs Theory, McGregor's Theory X & Y, Herzberg's Motivation-Hygiene Theory, Alderfer's ERG Theory, McClelland's Theory of Needs, Vroom's Expectancy Theory.

Module 3 [10L]

Group Behaviour: Characteristics of Group, Types of Groups, Stages of Group Development, Group Decision Making. Communication: Communication Process, Direction of Communication, Barriers to Effective Communication. Leadership: Definition, Importance, Theories of Leadership Styles.

Module 4 [8L]

Organizational Politics: Definition, Factors contributing to Political Behaviour. Conflict

Management: Traditional vis-a-vis Modern View of Conflict, Functional and Dysfunctional Conflict, Conflict Process, Negotiation – Bargaining Strategies, Negotiation Process. Organizational Design: Various Organizational Structures and their Effects on Human Behaviour, Concepts of Organizational Climate and Organizational Culture.

Text Books and/or Reference Material

1. Robbins, S. P. & Judge, T.A.: Organizational Behavior, Pearson Education, 15th Edn.
2. Luthans, Fred: Organizational Behavior, McGraw Hill, 12th Edn.
3. Shukla, Madhukar: Understanding Organizations – Organizational Theory & Practice in India, PHI
4. Fincham, R. & Rhodes, P.: Principles of Organizational Behaviour, OUP, 4th Edn.
5. Hersey, P., Blanchard, K.H., Johnson, D.E.- Management of Organizational Behavior Leading Human Resources, PHI, 10th Edn.

Course Name: Environmental Sciences

Course Code: MC401

Semester: 4

Contact Hours: L-T-P: 2-1-0

Credits: 0

Module 1 [5L]

Basic ideas of environment, basic concepts, man, society & environment, their interrelationship Mathematics of population growth and associated problems, Importance of population study in environmental engineering, definition of resource, types of resource, renewable, non-renewable, potentially renewable, effect of excessive use vis-à-vis population growth, Sustainable Development. Materials balance: Steady state conservation system, steady state system with non conservative pollutants, step function. Environmental degradation: Natural environmental Hazards like Flood, earthquake, Landslide-causes, effects and control/management; Anthropogenic degradation like Acid rain-cause, effects and control. Nature and scope of Environmental Science and Engineering. Elements of ecology: System, open system, closed system, definition of ecology, species, population, community, definition of ecosystem- components types and function. Structure and function of the following ecosystem: Forest ecosystem, Grassland ecosystem, Desert ecosystem, Aquatic ecosystems, Mangrove ecosystem (special reference to Sundar ban); Food chain [definition and one example of each food chain], Food web. Biogeochemical Cycle- definition, significance, flow chart of different cycles with only elementary reaction [Oxygen, carbon, Nitrogen, Phosphate, Sulphur]. Biodiversity- types, importance, Endemic species, Biodiversity Hot-spot, Threats to biodiversity, Conservation of biodiversity.

Module 2 [7L]

Atmospheric Composition: Troposphere, Stratosphere, Mesosphere, Thermosphere, Tropopause and Mesopause. Energy balance: Conductive and Convective heat transfer, radiation heat transfer, simple global temperature model [Earth as a black body, earth as

albedo], Problems. Green house effects: Definition, impact of greenhouse gases on the global climate and consequently on sea water level, agriculture and marine food. Global warming and its consequence, Control of Global warming. Earth's heat budget. Lapse rate: Ambient lapse rate Adiabatic lapse rate, atmospheric stability, temperature inversion (radiation inversion). Atmospheric dispersion: Maximum mixing depth, ventilation coefficient, effective stack height, smokestack plumes and Gaussian plume model. Definition of pollutants and contaminants, Primary and secondary pollutants: emission standard, criteria pollutant. Sources and effect of different air pollutantsSuspended particulate matter, oxides of carbon, oxides of nitrogen, oxides of sulphur, particulate, PAN. Smog, Photochemical smog and London smog. Depletion Ozone layer: CFC, destruction of ozone layer by CFC, impact of other green-house gases, effect of ozone modification. Standards and control measures: Industrial, commercial and residential air quality standard, control measure (ESP. cyclone separator, bag house, catalytic converter, scrubber (ventury), Statement with brief reference).

Module 3 [7L]

Hydrosphere, Hydrological cycle and Natural water. Pollutants of water, their origin and effects: Oxygen demanding wastes, pathogens, nutrients, Salts, thermal application, heavy metals, pesticides, volatile organic compounds. River/Lake/ground water pollution: River: DO, 5-day BOD test, Seeded BOD test, BOD reaction rate constants, Effect of oxygen demanding wastes on river [deoxygenation, reaeration], COD, Oil, Greases, pH. Lake: Eutrophication [Definition, source and effect]. Ground water: Aquifers, hydraulic gradient, ground water flow (Definition only) Standard and control: Waste water standard [BOD, COD, Oil, Grease], Water Treatment system [coagulation and flocculation, sedimentation and filtration, disinfection, hardness and alkalinity, softening] Waste water treatment system, primary and secondary treatments [Trickling filters, rotating biological contractor, Activated sludge, sludge treatment, oxidation ponds] tertiary treatment definition. Water pollution due to the toxic elements and their biochemical effects: Lead, Mercury, Cadmium, and Arsenic

Module 4 [5L]

Lithosphere; Internal structure of earth, rock and soil Solid Waste: Municipal, industrial, commercial, agricultural, domestic, pathological and hazardous solid wastes; Recovery and disposal method- Open dumping, Land filling, incineration, composting, recycling. Solid waste management and control (hazardous and biomedical waste). Definition of noise, effect of noise pollution, noise classification [Transport noise, occupational noise, neighbourhood noise] Definition of noise frequency, noise pressure, noise intensity, noise threshold limit value, equivalent noise level, L10 (18hr Index) ,n Ld.Noise pollution control. Environmental impact assessment, Environmental Audit, Environmental laws and protection act of India, Different international environmental treaty/ agreement/ protocol.

Text Books and/or Reference Material

1. M.P. Poonia & S.C. Sharma, Environmental Studies, Khanna Publishing House (AICTE Recommended Textbook – 2018)
2. Masters, G. M., "Introduction to Environmental Engineering and Science", Prentice-Hall of India Pvt. Ltd.,1991.
3. De, A. K., "Environmental Chemistry", New Age International

Course Name: Essential Studies for Professionals - IV

Course Code: HSMC402

Semester: 4

Contact Hours: L-T-P: 2-0-0

Credits: 0.5

Prerequisite: Basic Social Science from primary to high school, NCERTs

Module 1 [10L]

Constitution of India: Central Legislative System of India, State Legislative System of India, Constitutional Amendments.

Module 2 [12L]

History: Islam and Early Muslim Invaders, Delhi Sultanate, Rise of the Mughals (Till Akbar)

Module 3 [12L]

Geography: Rivers of India, Multipurpose River Valley Projects, Thermal Power Projects.

Module 4 [9L]

Economics: Tax System of India, Balance of Payment, Industrial Reforms, NITI Aayog and its relationship with 5 year plan, SDG

Module 5 [5L]

Current affairs and Static GK: Monthly Current Affairs Magazine

Text Books

1. Indian Constitution- M. Laxmikant
2. Indian Economy-Ramesh Singh
3. History of Modern India- Bepan Chandra
4. Geography of India- Majid Hussain

Reference Books

1. Current Affairs Magazine of IEM-UEM
2. Lucent GK

Course Name: Computer Organization & Architecture Lab

Course Code: PCCCS492

Semester: 4

Contact Hours: L-T-P: 0-0-4

Credits: 2
Prerequisite: Digital Electronics Lab

Module 1 [6L]
1. HDL introduction
2. Basic digital logic base programming with HDL

Module 2 [5L]
3. 8-bit Addition, Multiplication, Division
4. 8-bit Register design

Module 3 [5L]
5. Memory unit design and perform memory operations.
6. 8-bit simple ALU design

Module 4 [6L]
7. 8-bit simple CPU design
8. Interfacing of CPU and Memory

Course Name: Operating Systems Lab

Course Code: PCCCS493

Semester: 4

Contact Hours: L-T-P: 0-0-4

Credits: 2
Prerequisite: Data Structure & Algorithms Lab

Module1 [4L]

Managing Unix/Linux Operating System Creating a bash shell script, making a script executable, shell syntax (variables, conditions, control structures, functions, commands). Partitions, Swap space, Device files, Raw and Block files, Formatting disks, Making file systems, Superblock, I nodes, File system checker, Mounting file systems, Logical Volumes, Network File systems, Backup schedules and methods Kernel loading, init and the inittab file, Run-levels, Run level scripts. Password file management, Password security, Shadow file, Groups and the group file, Shells, restricted shells, user-management commands, homes and permissions, default files, profiles, locking accounts, setting passwords, Switching user, Switching group, Removing users & user groups.

Module2 [6L]

Process Starting new process, replacing a process image, duplicating a process image, waiting for a process, zombie process. Signal signal handling, sending signals, signal interface, signal sets.

Module3 [10L]

Semaphore Programming with semaphores (use functions semctl, semget,semop, set_semvalue, del_semvalue, semaphore_p, semaphore_v).

Module4 [10L]

POSIX Threads Programming with pthread functions (viz. pthread_create, pthread_join, pthread_exit, pthread_attr_init, pthread_cancel) Inter-process communication pipes(use functions pipe, popen, pclose), named pipes(FIFOs, accessing FIFO), message passing & shared memory(IPC version V).

Course Name: Design & Analysis of Algorithm Lab

Course Code: PCCCS494

Semester: 4

Contact Hours: L-T-P: 0-0-4

Credits: 2
Prerequisite: Data Structure & Algorithms Lab

Module 1 [8L]
Divide and Conquer:
1. Implement Binary Search using Divide and Conquer approach
2. Implement Merge Sort using Divide and Conquer approach
3. Implement Quick Sort using Divide and Conquer approach
4. Find Maximum and Minimum element from an array of integers using Divide and Conquer approach
Dynamic Programming:
5. Implement Matrix Chain Multiplication
6. Implement all pair of Shortest path for a graph (Floyd- Warshall Algorithm)
7. Implement 0/1 Knapsack Problem

Module 2 [6L]
Branch and Bound:
8. Implement 15 Puzzle Problem
Backtracking:
9. Implement 8-Queen problem

10. Graph Coloring Problem Hamiltonian Problem

Module 3 [6L]

Greedy Method:

11. Knapsack Problem Job sequencing with deadlines
12. Minimum Cost Spanning Tree by Prim's Algorithm Minimum Cost Spanning Tree by Kruskal's Algorithm

Module 4 [4L]

Graph Traversal Algorithm:

13. Implement Breadth First Search (BFS). Implement Depth First Search (DFS)
14. Implement Single Source shortest Path for a graph (Dijkstra , Bellman Ford Algorithm)

Course Name: Artificial Intelligence & Machine Learning Lab

Course Code: PCCCS495

Semester: 4

Contact Hours: L-T-P: 0-0-4

Credits: 1

Prerequisite: Data Structure & Algorithms Lab

Module 1 [6L]

Lab 1:

To implement Depth first search, Breadth first search

Lab 2:

To implement A* search

Module 2 [9L]

Lab 3:

Implement Perceptron. and to write a program to implement AND OR gates using Perceptron.

Lab 4:

Implementation of Decision Tree Classifier.

Lab 5:

Implementation of Random Forest Classifier.

Lab 6:

To study Long Short-Term Memory for Time Series Prediction

Module 3 [6L]

Lab 7:

To study Convolutional Neural Network and Recurrent Neural Network

Lab 8:

To study ImageNet, Google Net, ResNet convolutional Neural Networks

Module 4 [9L]

Lab 9:

To implement K-Mean's algorithm
Lab 10:
To implement K-Nearest Neighbour algorithm
Lab 11:
To implement Naïve Bayesian Classifier

Course Name: Skill Development for Professionals - IV

Course Code: HSMC482

Semester: 4

Contact Hours: L-T-P: 2-0-0

Credits: 0.5

Prerequisite: Basic Mathematics, General English from primary to high school.

Module 1 [6L]

Quantitative Aptitude Permutation & Combination, Probability, Geometry, Mensuration

Module 2 [12L]

Logical Reasoning
1) Seating Arrangement
a) Circular seating arrangement
b) Square seating Arrangement
c) Line Arrangement
2) Calendar And Clock
3) Miscellaneous Problems

Module 3 [12 L]

Verbal English
1) Sentence Corrections
2) Fill the blanks with appropriate words/articles/preposition/verbs/adver bs/conjunction.
3) Reading Comprehension (Advance Level)
4) Vocabulary

Learning Resources:
Text Books:
1. Objective General English- S.P Bakshi
2. English Grammar and Competition-S.C Gupta
3. Fast Track Objective Arithmetic- Rajesh Verma

4. Advance Maths- Rakesh Yadav
5. Verbal and Non-Verbal Reasoning- R.S Agarwal
6. A new approach to Reasoning- BS Sijwali
7. Quantitative Aptitude-R.S Agarwal

Semester V (Third year] Curriculum								
Sl. No.	**Type of course**	**Code**	**Course Title**	**Hours per week**				**Credits**
				Lecture	**Tutorial**	**Practical**	**Sessional**	
Theory Papers								
1	Engineering Science Course	ESC501	Signals & Systems	3	0	0	0	3
2	Professional Core Courses	PCC-CS501	Database Management Systems	3	0	0	0	3
3	Professional Core Courses	PCC-CS502	Formal Language & Automata Theory	3	0	0	0	3
4	Professional Core Courses	PCC-CS503	Object Oriented Programming	2	0	0	0	2
5	Professional Core Courses	PCC-CS504	Software Engineering	2	0	0	0	2
6	Humanities &Social Sciences including Management courses	HSMC-501	Humanities II (Principles of Management)	3	0	0	0	3
7	Professional Elective courses	PEC-CS-501	Professional Elective-I E. Soft Computing F. Computer Graphics	3	0	0	0	3
8	Mandatory Courses	MC501	Constitution of India/ Essence of Indian Knowledge Tradition	0	0	0	0	0
9	Humanities and social sciences including Management	HSMC502	Essential Studies for Professionals - V	2	0	0	0	0.5
	Total			**21**	**0**	**0**	**0**	**19.5**
Practical Papers								
1	Professional Core Courses	PCC-CS591	Database Management Systems Lab	0	0	4	0	2
2	Professional Core Courses	PCC-CS593	Object Oriented Programming Lab	0	0	4	0	2
3	Professional Core Courses	PCC-CS594	Software Engineering Lab	0	0	2	0	1
	Total			**0**	**0**	**10**	**0**	**5**
Sessional Papers								
1	Humanities and social sciences including Management	HSMC582	Skill Development for Professionals - V	0	0	0	2	0.5
2	Innovative Project	PROJ –CS501	Innovative Project - III	0	0	0	0	1
3	Mandatory Additional Requirements (MAR)	MAR581	Mandatory Additional Requirements (MAR)-V	0	0	0	0	0
4	MOOCs (Mandatory for Honours)	MOOCs 521	Massive Open Online Course 5.1 (Mandatory for B.Tech (Honours))	0	0	0	0	0
5	IFC			0	0	0	0	0
Total				**0**	**0**	**0**	**2**	**1.5**
Total				**21**	**0**	**10**	**2**	**26**

Course Code : ESC501
Course Title : Signals & Systems
Credit : 3

Course Outcome

On completion of the course, student will be able to

CO1.	Understand mathematical description and representation of continuous and discrete time signals and systems with the understanding of input output relationship for linear shift invariant system and the convolution operator for continuous and discrete time system.
CO2.	Understand and resolve the signals in frequency domain using Fourier series and Fourier transform.
CO3.	Understand the limitations of Fourier transform and need for Laplace transform and develop the ability to analyze the system in s- domain also understand the z-domain analysis of a system
CO4.	Understand the the Sampling Theorem and its implications

Detailed Syllabus

Module I **10L**

Signals and systems as seen in everyday life, and in various branches of engineering and science. Energy and power signals, continuous and discrete time signals, continuous and discrete amplitude signals. System properties: linearity: additivity and homogeneity, shift- invariance, causality, stability, realizability.
Linear shift-invariant (LSI) systems, impulse response and step response, convolution, input-output behavior with aperiodic convergent inputs. Characterization of causality and stability of linear shift invariant systems. System representation through differential equations.

Module II **8L**

Periodic and semi-periodic inputs to an LSI system, the notion of a frequency response and its relation to the impulse response, Fourier series representation, the Fourier Transform, convolution/multiplication and their effect in the frequency domain, magnitude and phase response, Fourier domain duality. The Discrete-Time Fourier Transform (DTFT) and the Discrete Fourier Transform (DFT). Parseval's Theorem. The idea of signal space and orthogonal bases.

Module III **8L**

Evolution of Transforms: Fourier Transform, Laplace Transform , Z-transform (single sided and Double sided) .The Laplace Transform, notion of eigen functions of LSI systems, a basis of eigen functions, region of convergence, poles and zeros of system, , solution to differential equations and system behavior using Laplace Transformation. The z-Transform for discrete time signals and systems- eigen functions, region of convergence, z-domain analysis.

Module IV **4L**

The Sampling Theorem and its implications- Spectra of sampled signals. Reconstruction: ideal interpolator, zero-order hold, first-order hold, and so on. Aliasing and its effects. Relation between continuous and discrete time systems.

Text/Reference books:

1. A.V. Oppenheim, A.S. Willsky and I.T. Young, "Signals and Systems", Prentice Hall,
   1983.
2. R.F. Ziemer, W.H. Tranter and D.R. Fannin, "Signals and Systems - Continuous and Discrete", 4th edition, Prentice Hall, 1998.
3. Papoulis, "Circuits and Systems: A Modern Approach", HRW, 1980

Course Code **:** PCC-CS501
Course Title **:** Database Management Systems
Credit **:** 3

Course Outcome

On completion of the course, student will be able

CO1.	To understand the different issues involved in the design and implementation of a database.
CO2.	To understand the relational database design principles.
CO3.	To be familiar with the basic issues of transaction processing and concurrency control.
CO4.	To be familiar with basic database storage structures, access methodologies and query processing techniques involved in query optimization.

SYLLABUS:

Module I (Mapped to CO1)

1. *Introduction:* Concept & Overview of DBMS, Data Models, Database Languages, Database Administrator, Database Users, Three Schema architecture of DBMS.
2. *Entity-Relationship Model:* Basic concepts, Design Issues, Mapping Constraints, Keys, Entity-Relationship Diagram, Weak Entity Sets, Extended E-R features.

Module II (Mapped to CO2)

3. *Relational Model:* Structure of Relational Databases, Relational Algebra, Relational Calculus, Extended Relational Algebra Operations, Views, Modifications of the Database.
4. *SQL and Integrity Constraints*: Concept of DDL, DML, DCL. Basic Structure, Set operations, Aggregate Functions, Null Values, Domain Constraints, Referential Integrity Constraints, assertions, views, Nested Subqueries, Database security application development using SQL, Stored procedures and triggers.

5. *Relational Database Design:* Functional Dependency, Different anomalies in designing a Database., Normalization using functional dependencies, Decomposition, Boyce-Codd Normal Form, 3NF, Normalization using multi-valued dependencies, 4NF, 5NF

Module III (Mapped to CO3)

6. *Transaction Management System:* Transaction processing, Transaction States, Serializability Concurrency control and Recovery Management: Lock base protocols, two-phase locking, Timestamp based protocol, Log based recovery.

Module IV (Mapped to CO4)

7. *Internals of RDBMS:* Physical data structures, Query optimization: join algorithm, statistics, and cost-based optimization.
8. *File Organization & Index Structures:* File & Record Concept, placing file records on Disk, Fixed and Variable Sized Records, Types of Single-Level Index (primary, secondary, clustering), Multilevel Indexes, Dynamic Multilevel Indexes using B tree and B+ tree.

Text Books & Reference Books:

1. "Database System Concepts" 6th Edition - by Abraham Silberschatz, Henry F. Korth, S. Sudarshan, McGraw-Hill
2. ."Principles of Database and Knowledge – Base Systems", Vol 1 by J. D. Ullman Computer Science Press.
3. "Fundamentals of Database Systems" , 5th Edition by R. Elmasri and S. Navathe
4. Database Management Systems, R.P. Mahapatra, Khanna Publishing House

Course Code : PCC-CS502
Course Title : Formal Language & Automata Theory
Credit : 3

Course Outcome

On completion of the course,

CO1.	Students will be able to define mathematical model of a system and recognize the behaviour of a system.
CO2.	Students will be able to understand the regular expression.
CO3.	Students will be able to understand the concept of grammar and language and apply the knowledge in compiler design.
CO4.	Students would be able to understand the computability and complexity of a problem.

SYLLABUS:

Module I (Mapped to CO1)

9. Alphabet, languages and grammars, productions and derivation, Chomsky hierarchy of languages.

Module II (Mapped to CO2)

10. Regular expressions and languages, deterministic finite automata (DFA) and equivalence with regular expressions, nondeterministic finite automata (NFA) and equivalence with DFA, regular grammars and equivalence with finite automata, properties of regular languages, pumping lemma for regular languages, minimization of finite automata.

Module III (Mapped to CO3)

11. Context-free grammars (CFG) and languages (CFL), Chomsky and Greibach normal forms, nondeterministic pushdown automata (PDA) and equivalence with CFG, parse trees, ambiguity in CFG, pumping lemma for context-free languages, deterministic pushdown automata, closure properties of CFLs.

Module IV (Mapped to CO4)

12. Context-sensitive grammars (CSG) and languages, linear bounded automata and equivalence with CSG. Turing machines: The basic model for Turing machines (TM), Turing-recognizable (recursively enumerable) and Turing-decidable (recursive) languages and their closure properties, variants of Turing machines, nondeterministic TMs and equivalence with deterministic TMs, unrestricted grammars and equivalence with Turing machines, TMs as enumerators.

13. Church-Turing thesis, universal Turing machine, the universal and diagonalization languages, reduction between languages and Rice s theorem, undecidable problems about languages.

Text Books & Reference Books:

1. Harry R. Lewis and Christos H. Papadimitriou, Elements of the Theory of Computation, Pearson Education Asia.
2. Dexter C. Kozen, Automata and Computability, Undergraduate Texts in Computer Science, Springer.
3. Michael Sipser, Introduction to the Theory of Computation, PWS Publishing.
4. John Martin, Introduction to Languages and The Theory of Computation, Tata McGraw Hill.
5. John E. Hopcroft, Rajeev Motwani and Jeffrey D. Ullman, Introduction to Automata Theory, Languages, and Computation, Pearson Education Asia.
6. Theory of Computer Science: Automata, Languages and Computation, Mishra,Phi
7. Theory of Computation, RB Patel & Prem Nath, KhannaPublications

Course Code : PCC CS503
Course Title : Object Oriented Programming
Credit : 2

Course Outcome
On completion of the course,

CO1.	Students will be able to specify simple abstract data types and design implementations, using abstraction functions to document them.
CO2.	Students will be able to recognize features of object-oriented design such as encapsulation, polymorphism, inheritance, and composition of systems based on object identity.
CO3.	Students will be able to implement the ideas of MVC architecture regarding the development process of any software architecture.
CO4.	Students will be able to design applications with an event-driven graphical user interface.

Module No.	Content	Hrs./Unit	CO
1	Abstract data types and their specification. How to implement an ADT. Concrete state space, concrete invariant, abstraction function. Implementing operations, illustrated by the Text example.	8	1
2	Features of object-oriented programming. Encapsulation, object identity, polymorphism – but not inheritance. Inheritance in OO design. Design patterns. Introduction and classification. The iterator patterns.	14	2
3	Model-view-controller pattern. Commands as methods and as objects. Implementing OO language features. Memory management.	6	3
4	Generic types and collections GUIs. Graphical programming with Scale and Swing. The software development process	6	4

Course Code : PCC CS504
Course Title : Software Engineering
Credit : 2

Course Outcome
On completion of the course,

CO1.	Students will be able to decompose the given project in various phases of a life cycle as well as various cost-benefit analysis.
CO2.	Students will be able to choose appropriate process model depending on the user requirements and implement proper design, coding and testing methodologies.
CO3.	Students will be able to know various processes used in all the phases of project management.
CO4.	Students can apply the knowledge of various static and dynamic models and UML diagrams on account of project development.

Module No.	Content	Hrs./Unit	CO
1	Overview of System Analysis & Design, Business System Concept, System Development Life Cycle, Waterfall Model, Spiral Model, Feasibility Analysis, Technical Feasibility, Cost- Benefit Analysis, COCOMO model.	10	1
2	System Design – Context diagram and DFD, Problem Partitioning, Top-Down and Bottom-Up design; Decision tree, decision table and structured English; Functional vs. Object- Oriented approach. Coding & Documentation – Structured Programming, OO Programming, Information Hiding, Reuse, System Documentation. Testing – Levels of Testing, Integration Testing, Test case Specification, Reliability Assessment, Validation & Verification, Metrics, Monitoring & Control.	17	2
3	Software Project Management – Project Scheduling, Staffing, Software Configuration Management, Quality Assurance, Project Monitoring.	8	3
4	Static and dynamic models, why modeling, UML diagrams: Class diagram, interaction diagram: collaboration diagram, sequence diagram, state chart diagram, activity diagram, implementation diagram.	7	4

Course Code : HSMC501
Course Title : Humanities – II (Principles of Management)

Credit : 3

Course Outcome
On completion of the course, students will be able to

CO1.	Understand basics of Management.
CO2.	Understanding Management in the context of Society & Leadership
CO3.	Use of Decision making tools and understanding operations and production management
CO4.	Understanding Customer management

SYLLABUS:

Module I:

1. Basic concepts of management: Definition – Essence, Functions, Roles, Level.
   Functions of Management: Planning – Concept, Nature, Types, Analysis, Management by objectives; Organisation Structure – Concept, Structure, Principles, Centralization, Decentralization, Span of Management; Organisational Effectiveness.

Module II:

2. Management and Society – Concept, External Environment, CSR, Corporate Governance, Ethical Standards.People Management – Overview, Job design, Recruitment & Selection, Training & Development, Stress Management.
   Managerial Competencies – Communication, Motivation, Team Effectiveness, Conflict Management, Creativity, Entrepreneurship.
   Leadership: Concept, Nature, Styles.

Module III:

3. Decision making: Concept, Nature, Process, Tools & techniques.
   Economic, Financial & Quantitative Analysis – Production, Markets, National Income Accounting, Financial Function & Goals, Financial Statement & Ratio Analysis,
   Quantitative Methods – Statistical Inference, Forecasting, Regression Analysis, Statistical Quality Control.

Module IV:

4. Customer Management – Market Planning & Research, Marketing Mix, Advertising & Brand Management.Operations & Technology Management – Production & Operations Management, Logistics & Supply Chain Management, TQM, Kaizen & Six Sigma, MIS.

Readings:

1. Management: Principles, Processes & Practices – Bhat, A & Kumar, A (OUP).

2. Essentials for Management – Koontz, Revised edition, Tata McGraw Hill (TMH)
3. Management – Stoner, James A. F. (Pearson)
4.Management - Ghuman, Tata McGraw Hill(TMH)

Course Code : PECCS 501F
Course Title : Computer Graphics
Credit : 3

COURSE OUTCOMES
CO1: Define computer graphics and graphics systems
CO2: Understand raster and vector graphics devices, light and color models
CO3: Analyze different algorithms for line drawing, circle drawing, polygon filling etc.
CO4: Apply geometrical transformations, viewing and clipping algorithms on 2D and 3D objects
CO5: Evaluate different curve generation algorithms, removal of hidden surfaces, shading models
CO6: Creating a typical graphics using Curve representation, 3D transformations

Syllabus:

Introduction to computer graphics & graphics systems [6L]: Overview of computer graphics, representing pictures, preparing, presenting & interacting with pictures for presentations; Visualization & image processing; RGB color model, direct coding, lookup table; storage tube
graphics display, Raster scan display, 3D viewing devices, Plotters, printers, digitizers, Light pens
etc.; Active & Passive graphics devices; Computer graphics software.
Scan conversion [8L]: Points & lines, Line drawing algorithms; DDA algorithm, Bresenham's line
algorithm, Circle generation algorithm; Ellipse generating algorithm; scan line polygon, fill algorithm,
boundary fill algorithm, flood fill algorithm.
2D transformation & viewing [15L]: Basic transformations: translation, rotation, scaling; Matrix
representations & homogeneous coordinates, transformations between coordinate systems; reflection
shear; Transformation of points, lines, parallel lines, intersecting lines. Viewing pipeline, Window to
view port co-ordinate transformation, clipping operations, point clipping, line clipping, clipping
circles, polygons & ellipse. Cohen and Sutherland line clipping, Sutherland-Hodgeman Polygon
clipping
3D transformation & viewing [5L]: 3D transformations: translation, rotation, scaling & other transformations. Rotation about an arbitrary axis in space, reflection through an arbitrary plane;
general parallel projection transformation; clipping, view port clipping, 3D viewing.
Curves [3L]: Curve representation, surfaces, designs, Bezier curves, B-spline curves, end conditions

for periodic B-spline curves, rational B-spline curves.
Hidden surfaces [3L]: Depth comparison, Z-buffer algorithm, Back face detection, BSP tree method,
the Painter's algorithm, scan-line algorithm; Hidden line elimination, wire frame methods , fractal -
geometry.
Color & shading models [2L]: Light & color model; interpolative shading model; Texture.

Books:
1. Hearn, Baker – "Computer Graphics (C version 2nd Ed.)" – Pearson education
2. Z. Xiang, R. Plastock – " Schaum's outlines Computer Graphics (2nd Ed.)" – TMH
3. D. F. Rogers, J. A. Adams – "Mathematical Elements for Computer Graphics (2nd Ed.)" – TMH

Soft Computing
Code: PEC-CS501E
Contacts: 3L

CO1: Basic concepts of different aspects of Soft Computing
CO2: Understanding and interpreting various conceptual aspects of Soft Computing
CO3: Implementation of the concepts of Soft Computing
CO4: Analysing various concepts of Soft Computing
CO5: Evaluation of various concepts of Soft Computing
CO6: Generation models using various concepts of Soft Computing

Text book and Reference books:
1. Principles of Soft Computing , S N Sivanandam, S. Sumathi, John Wiley & Sons
2. Fuzzy logic with engineering applications, Timothy J. Ross, John Wiley and Sons.
3. Fuzzy Sets and Fuzzy Logic: Theory and Applications, George J. Klir and Bo Yuan, Prentice Hall
4. Neuro-Fuzzy and Soft computing, Jang, Sun, Mizutani, PHI
5. Neural Networks: A Comprehensive Foundation (2nd Edition), Simon Haykin, Prentice Hall.
6. Neural Networks, Fuzzy Logic and Genetic Algorithms, S. Rajasekaran and G. A. V. Pai, PHI
7. Multi-objective optimization using evolutionary algorithms, Kalyanmoy Deb, Wiley
8. Genetic Algorithms in search, Optimization & Machine Learning by David E. Goldberg
9. Evolutionary Optimization Algorithms Dan Simon, John Wiley & Sons

		CO	BL
Introduction to Soft Computing	Concept of computing systems.	1	1
	"Soft" computing vs. "Hard" computing	1	1
	Characteristics of Soft computing	1	1
	Some applications of Soft computing techniques	2	2
Fuzzy logic	Introduction to Fuzzy logic.	1	1
	Fuzzy sets and membership functions.	2	2
	Operations on Fuzzy sets.	2	2
	Fuzzy relations, rules, propositions, implications and inferences.	3	3
	Defuzzification techniques.	4	4
	Fuzzy Inference System- Mamdani Fuzzy Models – Sugeno Fuzzy Models	5	5
	Some applications of Fuzzy logic.	6	6
Genetic Algorithms	Concept of "Genetics" and "Evolution" and its application to probabilistic search techniques	2	2
	Basic GA framework and different GA architectures.	3	3
	GA operators: Encoding, Crossover, Selection, Mutation, etc.	4	4
	Solving single-objective optimization problems using GAs.	5	5
Multi-objective Optimization Problem Solving	Concept of multi-objective optimization problems (MOOPs) and issues of solving them.	2	2
	Multi-Objective Evolutionary Algorithm (MOEA).	3	3
	Non-Pareto approaches to solve MOOPs	4	4
	Pareto-based approaches to solve MOOPs	5	5
	Some applications with MOEAs.	6	6
Artificial Neural Networks	Biological neurons and its working.	2	2
	Simulation of biological neurons to problem solving.	3	3
	Different ANNs architectures.	4	4
	Training techniques for ANNs.	5	5
	Applications of ANNs to solve some real life problems.	6	6
Swarm intelligence	ant colony optimization, ant-based routing	6	6
	swarm intelligence in bees,	6	6
	flocks of birds,	6	6
	shoals of fish,	6	6
	particle swarm optimization	6	6

Course Code : MC501
Course Title : Constitution of India & Essence of Indian Knowledge Tradition
Credit : 0

Course Outcome

CO1.	Describe different features of Indian constitution.
CO2.	Power and functioning of Union, state and local self-government. Structure, jurisdiction and function of Indian Judiciary. Basics of PIL and guideline for admission of PIL.
CO3.	Functioning of local administration starting from block to Municipal Corporation.
CO4.	Identify authority to redress a problem in the profession and in the society.

SYLLABUS:

Module I:

Indian Constitution: Sources and constitutional history, Features: Citizenship, Preamble, Fundamental Rights and Duties, Directive Principles of State Policy.

Module II:

Union government and its administration: Structure of the Indian Union: Federalism, Centre- State relationship, President: Role, power and position, PM and Council of ministers, Cabinet and Central Secretariat, Lok Sabha, Rajya Sabha. State government and its administration: Governor: Role and Position, CM and Council of ministers, State Secretariat: Organisation, Structure and Functions.

Module III:

Supreme court: Organization of supreme court, procedure of the court, independence of the court, jurisdiction and power of supreme court. High court: Organization of high court, procedure of the court, independence of the court, jurisdiction and power of supreme court. Subordinate courts: constitutional provision, structure and jurisdiction. National legal services authority, Lok adalats, family courts, gram nyayalays. Public interest litigation (PIL): meaning of PIL, features of PIL, scope of PIL, principle of PIL, guidelines for admitting PIL

Module IV:

Local Administration: District's Administration head: Role and Importance, Municipalities: Introduction, Mayor and role of Elected Representative, CEO of Municipal Corporation, Pachayati raj: Introduction, PRI: Zila Pachayat, Elected officials and their roles, CEO Zila Pachayat: Position and role, Block level: Organizational Hierarchy (Different departments), Village level: Role of Elected and Appointed officials, Importance of grass root democracy.

Learning Resources:

1. Indian polity, M, Laxmikanth, MC Graw Hill education, 5th Edition.
2. DD Basu, "Introduction to the constitution of India", 21st Edition, Lexis Nexis Books Publication ltd, India

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Course Code	**:**	HSMC 502
Course Title	**:**	ESP-V
Credit	**:**	2
Pre-Requisites	**:**	Basics of Digital Electronic, Programming, Discrete Mathematics and Computer Organization
L-T-P	:	2-0-0 (Total Contact Hrs. 2)

Course Objective:

1. To learn about basic of digital circuits for professional exams
2. To learn about fundamentals of computer programming for various exams
3. To learn about basic of discrete mathematics for professional exams
4. To learn about fundamentals of computer organizations for various exams

Course Outcome
At the end of the course the students will be able

CO1.	To develop an understanding of Digital electronic circuit components and their working principles.
CO2.	To learn all types of linear, non-linear data structures and calculate time complexity and space complexity of any given algorithm.
CO3.	To understand memory technology and communication among processing elements.
CO4.	To use logical notation & Perform logical proofs, recursive functions and solve recurrence relations and principles of counting

Course Content:

Module No.	Description	Hours
1.	**Digital Logic:** Digital Logic Boolean algebra. Combinational and sequential circuits. Minimization. Number representations and computer arithmetic (fixed and floating point).	6
2.	**Programming and Data Structures** Programming and Data Structures Programming in C. Recursion. Arrays, stacks.	12
3.	**Computer Organization and Architecture:** Machine instructions and addressing modes. ALU, data path and control unit. Instruction pipelining.	12
4.	**Discrete Mathematics:** Propositional and first order logic. Sets, relations, functions, partial orders and lattices. Groups. Graphs: connectivity, matching, coloring. Combinatorics: counting, recurrence relations, generating functions.	6

Learning Resources:

Ref. Books:

1. G.K publishers GATE Computer Science Engineering,

2. McGraw hill GATE 2020 Computer Science Engineering,
3. Wiley GATE 2020 Computer Science Engineering,

Course Code : PCC-CS591
Course Title : Database Management Systems Lab
Credit : 2

Course Outcome

CO1.	Apply the basic concepts of Database Systems and Applications.
CO2.	Use the basics of SQL and construct queries using SQL in database creation and interaction.
CO3.	Design a commercial relational database system (Oracle) by writing SQL using the system.
CO4.	Analyze and Select storage and recovery techniques of database system.

SYLLABUS:

Module I (Mapped to CO1)

Experiment 1
CODD'S Rules, Student should decide on a case study and formulate the problem statement.
Experiment 2
Conceptual Designing using ER Diagrams (Identifying entities, attributes, keys and relationships between entities, cardinalities, generalization, specialization etc.)
Note: Student is required to submit a document by drawing ER Diagram to the Lab teacher.
Experiment 3
Converting ER Model to Relational Model (Represent entities and relationships in Tabular form, Represent attributes as columns, identifying keys)
Note: Student is required to submit a document showing the database tables created from ER Model.

Module II (Mapped to CO2)

Experiment 4
Normalization -To remove the redundancies and anomalies in the above relational tables, Normalize up to Third Normal Form
Experiment 5
Creation of Tables using SQL- Overview of using SQL tool, Data types in SQL, Creating Tables (along with Primary and Foreign keys), Altering Tables and Dropping Tables
Experiment 6
Practicing DML commands- Insert, Select, Update, Delete, GROUP BY Queries, SUB Queries, CUBE, ROLLUP Functions, WITH, CASE Operators.

Module III (Mapped to CO3)

Experiment 7
Practicing Queries using ANY, ALL, IN, EXISTS, NOT EXISTS, UNION, INTERSECT,

Oracle Integrity Constraints (PRIMARY KEY, NOT NULL, CHECK, FOREIGN KEY, UNIQUE)

Default Values, Dropping Constraints, Disabling and Enabling Constraints, Deferring Constraints Check Viewing Information about Constraints etc.

Experiment 8
Practicing Sub queries (Nested, Correlated) and Joins (Inner, Outer and Equi).
Oracle Transaction Control Language (COMMIT, ROLLBACK,SAVEPOINT) Data Control Language (GRANT, REVOKE)

Experiment 9
Practice Queries using COUNT, SUM, AVG, MAX, MIN, GROUP BY, HAVING, VIEWS Creation and Dropping.
Oracle SQL Functions

Number Functions (Math Functions), Character Functions, Miscellaneous Functions, Aggregate Functions, Date and Time Functions

Module IV (Mapped to CO4)
Experiment 10

Practicing on Triggers - creation of trigger, Insertion using trigger, Deletion using trigger, Updating using trigger

Experiment 11

Procedures- Creation of Stored Procedures, Execution of Procedure, and Modification of Procedure.

Overview of PL/SQL Elements

Blocks, Variables and Constants, Using SQL in PL/SQL, Branching and Conditional Control, Looping Statements, GOTO, Procedures, Functions and Packages Records, Object Types, Collections, Triggers, Error Handling

Experiment 12

Cursors- Declaring Cursor, Opening Cursor, Fetching the data, closing the cursor.

Text Books & Reference Books:

1. Oracle Database 12C: The Complete Reference
2. Database Management System (DBMS)A Practical Approach, Rajiv Chopra S. Chand Publishing

Course Code : PCC-CS593
Course Title : Object Oriented Programming Lab
Credit : 2

Course Outcome	
CO1.	Students should be able to apply Encapsulation and basics of ava programming language
CO2.	Students should be able to apply Inheritance and polymorphism with Java programming language
CO3.	Students should be able to demonstrate String handling features of Java programming language
CO4.	Students should be able to apply GUI and Event driven programming using Java programming language

SYLLABUS:

Module	Description
1	Assignments on class, constructor, Variables, static variables, reference variables, arrays, wrapper class, Vectors,
2	Assignments on inheritance, overloading, overriding, developing interfaces- multiple inheritance, extending interfaces, creating and accessing packages
3	Assignments on String handling
4	Assignments on applet programming

Course Code : PCC-CS594
Course Title : Software Engineering Lab
Credit : 1

Course Outcomes	
CO1.	Ability to analysis and design of complex systems and meet ethical standards, legal responsibilities

CO2.	Ability to apply software engineering principles, techniques and develop, maintain, evaluate large-scale software systems
CO3.	To produce efficient, reliable, robust and cost-effective software solutions and perform independent research and analysis
CO4.	Ability to work as an effective member or leader of software engineering teams and manage time, processes and resources effectively by prioritizing competing demands to achieve personal and team goals

SYLLABUS:

Module I: Problem Analysis and Project Planning -Thorough study of the problem – Identify Project scope, Objectives and Infrastructure.

Module II: Software Requirement Analysis – Describe the individual Phases/modules of the project and Identify deliverables. Identify functional and non-functional requirements.

Module III: Data Modeling – Use work products – data dictionary.

Module IV: Software Designing - Develop use case diagrams and activity diagrams, build and test class diagrams, sequence diagrams and add interface to class diagrams. Prototype model – Develop the prototype of the product.

Course Code : HSMC582
Course Title : SKILL DEVELOPMENT FOR PROFESSIONALS - V
Credit : 1
Pre-Requisites : Basic Mathematics, General English from primary to high school.
L-T-P : 2-0-0 (Total Contact Hrs. 2)

Course Objective:
1. To enhance the aptitude & analytical skill of students with multiple tricky approaches.
2. To prepare the students for various competitive examinations & professional exams

Course Outcomes	
CO1.	The ability to communicate effectively with a range of audiences.
CO2.	The ability to face the test and interview conducted by different companies and succeed
CO3.	The ability to recognize the need for continuing professional development.
CO4.	The ability to succeed in competitive exams

Course Content:

Module No.	Description	Hours	Blooms Level	PO(1..12) Mapping
1.	**Quantitative Aptitude & Data Interpretation**- Miscellaneous	12	L1 (Remember) L2 (Understand) L4 (Analyse)	PO1,PO2, PO10
2.	**Logical Reasoning** 1) Statement And Assumption, 2) Statement And Conclusion, 3) Statement And Course Of Action, 4) Cause And Effect, 5) Drawing Inference	12	L1 (Remember) L2 (Understand) L3 (Apply) L4 (Analyze)	PO1,PO2, PO10
3.	**Verbal English** **1) Sentence Corrections** 2) Fill the **blanks** with appropriate words/articles/preposition/verbs/adverbs/conjunction. 3) **Reading Comprehension** (Advance Level) **4) Vocabulary**	12	L1 (Remember) L2 (Understand) L3 (Apply) L4 (Analyze)	PO1,PO2, PO10

Learning Resources:

Reference Books:

1. Objective General English- S.P Bakshi
2. English Grammar and Competition-S.C Gupta
3. Fast Track Objective Arithmetic- Rajesh Verma
4. Advance Maths- Rakesh Yadav
5. Verbal and Non-Verbal Reasoning- R.S Agarwal
6. A new approach to Reasoning- BS Sijwali
7. Quantitative Aptitude-R.S Agarwal

Semester VI (Third year] Curriculum								
Sl. No.	**Type of course**	**Code**	**Course Title**	**Hours per week**				**Credits**
				Lecture	**Tutorial**	**Practical**	**Sessional**	
Theory Papers								
1	Professional Core Courses	PCC-CS601	Compiler Design	3	0	0	0	3
2	Professional Core Courses	PCC-CS602	Computer Networks	3	0	0	0	3
3	Professional Core Courses	PCC-CS603	Cloud Computing & IOT	2	0	0	0	2
4	Professional Elective courses	PEC-CS601	Professional Elective-II D: Parallel & Distributed System E: Neural Network & Deep Learning	3	0	0	0	3
5	Professional Elective courses	PEC-CS602	Professional Elective-III D: Data Mining and Analytics E: Image Processing and Pattern Recognition F: Social Media Analytics	3	0	0	0	3
6	Open Elective courses	OEC - CS601	Open Elective-I D: Entrepreneurship & Startups E. Financial & Cost Accounting	3	0	0	0	3
7	Humanities and social sciences including Management	HSMC602	Essential Studies for Professionals - VI	2	0	0	0	2
	Total			**19**	**0**	**0**	**0**	**19**
Practical Papers								
1	Professional Core Courses	PCC-CS691	Compiler Design Lab	0	0	4	0	2
2	Professional Core Courses	PCC-CS692	Computer Networks Lab	0	0	4	0	2
3	Professional Core Courses	PCC-CS693	Cloud Computing & IOT Lab	0	0	2	0	1
4	Innovative Project	PROJ-CS601	Project-1	0	0	6	0	3
	Total			**0**	**0**	**14**	**0**	**7**
Sessional Papers								

1	Humanities and social sciences including Management	HSMC682	Skill Development for Professionals - VI	0	0	0	2	1
2	Mandatory Additional Requirements (MAR)	MAR681	Mandatory Additional Requirements (MAR)-VI	0	0	0	0	0
3	MOOCs (Mandatory for Honours)	MOOCs 621	Massive Open Online Course 6.1 (Mandatory for B.Tech (Honours))	0	0	0	1	1
	Total			**0**	**0**	**0**	**4**	**3**
	Total			**19**	**0**	**14**	**4**	**29**

Course Code : PCC CS601
Course Title : Compiler Design
Credit : 3

Prerequisites: Basic Computation & Principles of Computer Programming, Data structures and algorithms, Formal Language & Automata Theory

Module 1 [9L]

Overview of compilation & Lexical Analysis- The structure of a compiler and applications of compiler technology; Lexical analysis - The role of a lexical analyzer, specification of tokens, recognition of tokens, hand-written lexical analyzers, LEX, examples of LEX programs.

Introduction to syntax analysis -Role of a parser, use of context-free grammars (CFG) in the specification of the syntax of programming languages, techniques for writing grammars for programming languages (removal left recursion, etc.), non- context-free constructs in programming languages, parse trees and ambiguity, examples of programming language grammars.

Module 2 [16L]

Top-down parsing- FIRST & FOLLOW sets, LL(1) conditions, predictive parsing, recursive descent parsing, error recovery. LR-parsing - Handle pruning, shift-reduce parsing, viable prefixes, valid items, LR(0) automaton, LR-parsing algorithm, SLR(1), LR(1), and LALR(1) parsing. YACC, error recovery with YACC and examples of YACC specifications.

Syntax-directed definitions (attribute grammars)-Synthesized and inherited attributes,

examples of SDDs, evaluation orders for attributes of an SDD, dependency graphs. S-attributed and L- attributed SDDs and their implementation using LR-parsers and recursive descent parsers respectively.

Module 3 [8L]

Semantic analysis- Symbol tables and their data structures. Representation of "scope". Semantic analysis of expressions, assignment, and control-flow statements, declarations of variables and functions, function calls, etc., using S- and L-attributed SDDs (treatment of arrays and structures included). Semantic error recovery

Intermediate code generation - Different intermediate representations –quadruples, triples, trees, flow graphs, SSA forms, and their uses. Translation of expressions (including array references with subscripts) and assignment statements. Translation of control-flow statements – it- then- else, while-do, and switch. Short-circuit code and control-flow translation of Boolean expressions. Back patching. Examples to illustrate intermediate code generation for all constructs

Module 4 [12L]

Run-time environments:- Stack allocation of space and activation records. Access to non-local data on the stack in the case of procedures with and without nesting of procedures.

Introduction to machine code generation and optimization- Simple machine code generation, examples of machine-independent code optimizations.

Code improvement, local optimization, global optimization, loop optimization, peep-hole optimization etc. Architecture dependent code improvement: instruction scheduling (for pipeline), loop optimization (for cache memory) etc. Register allocation and target code generation Advanced topics: Type systems, data abstraction, compilation of Object Oriented features and non-imperative programming languages.

Text Books:

1. Compilers: Principles, Techniques, and Tools , by A.V. Aho, Monica Lam, Ravi Sethi, and J.D. Ullman, (2nd ed.), Addison-Wesley, 2007 (main text book, referred to as ALSU in lab assignments).

2. K.D. Cooper, and Linda Torczon, Engineering a Compiler, Morgan Kaufmann, 2004.

Reference Books:

1. K.C. Louden, Compiler Construction: Principles and Practice, Cengage Learning, 1997.

2. D. Brown, J. Levine, and T. Mason, LEX and YACC, O"Reilly Media, 1992.

Course Code	**:**	PCC CS602
Course Title	**:**	Computer Networks
Credit	**:**	3

Module 1 [8L]

Data communication Components: Representation of data and its flow Networks , Various Connection Topology, Protocols and Standards, OSI model, Transmission Media, LAN: Wired LAN, Wireless LANs, Connecting LAN and Virtual LAN, Techniques for Bandwidth utilization: Multiplexing - Frequency division, Time division and Wave division, Concepts on spread spectrum. Network core; Delay, Loss and throughput in Packet-Switched Networks;

Module 2 [10L]

Data Link Layer and Medium Access Sub Layer: Error Detection and Error Correction - Fundamentals, Block coding, Hamming Distance, CRC; Flow Control and Error control protocols - Stop and Wait, Go back – N ARQ, Selective Repeat ARQ, Sliding Window, Piggybacking, Random Access, Multiple access protocols -Pure ALOHA, Slotted ALOHA, CSMA/CD,CDMA/CA

Module 3 [12L]

Network Layer: Switching, Logical addressing – IPV4, IPV6; Address mapping – ARP, RARP, BOOTP and DHCP–Delivery, Forwarding and Unicast Routing protocols. Transport Layer: Process to Process Communication, User Datagram Protocol (UDP), Transmission Control Protocol (TCP), SCTP Congestion Control; Quality of Service, QoS improving techniques: Leaky Bucket and Token Bucket algorithm.

Module 4 [6L]

Application Layer: Domain Name Space (DNS), DDNS, TELNET, EMAIL, File Transfer Protocol (FTP), WWW, HTTP, SNMP, Bluetooth, Firewalls, Basic concepts of Cryptography

Text Books:

1. Data Communication and Networking, 4th Edition, Behrouz A. Forouzan, McGrawHill.

2. Data and Computer Communication, 8th Edition, William Stallings, Pearson Prentice Hall India.

Reference Books:

1. Computer Networks, 8th Edition, Andrew S. Tanenbaum, Pearson New International Edition.

2. Internetworking with TCP/IP, Volume 1, 6th Edition Douglas Comer, Prentice Hall of India.

3. TCP/IP Illustrated, Volume 1, W. Richard Stevens, Addison-Wesley, United States of America.

Course Code: PCC- CS603
Course Title: Cloud Computing & IoT
Credit: 2
Pre-requisites:
Students should have prior knowledge of computer networks.

Course Outcome:
CO 1: Students are able to understand the various concepts of the IoT and their technologies.
CO 2: Students are able to develop the IoT application using different hardware platforms.
CO 3: Students are able to implement the various IoT Protocols and deploy the IoT application into a cloud environment.
CO 4: Students are able to understand the basic principles of cloud computing.

Syllabus:
Module 1: (6) Introduction to IoT – IoT definition – Characteristics – IoT Complete Architectural Stack – IoTenabling Technologies – IoT Challenges. M2M communication paradigm, Sensors, actuator concepts, Sensor networks, WSN, RFID, data gathering and sensor fusion. Sensors and Hardware for IoT – Hardware Platforms – Arduino, Raspberry Pi, Node MCU. Case study with any one of the boards and data acquisition from sensors. Python MQTT, MQTT Paho application development. Introduction to Node-Red IoT Platform.
Module 2: (6)

Protocols for IoT – Infrastructure protocol (IPV4/V6/RPL), Device Management Protocols. IoT protocols, fundamentals of Wireless network protocols and standards. Protocol stacks. MQTT, MQTT-SN, XMPP, COAP, GRPC, IoT streaming protocols. Sensor interfacing with various embedded devices, Light weight embedded computing platforms for IoT. IoT Security and Privacy. A Case Study with MQTT/CoAP usage-IoT privacy, security and vulnerability solutions. Case studies with architectural analysis: IoT applications – Smart City – Smart Water – Smart Agriculture. Advanced topics: IIoT, IoV, IoD, interoperability in IoT, Smart Grid systems, Smart cities application, Sensor cloud, device to cloud. Case study: 5G-IoT, Software defined networks, Edge, Fog and Dew computing.

Module 3: (6)

Defining Cloud Computing Paradigm. Cloud Types – NIST model, Cloud Cubemodel, Deployment models (Public, Private, Hybrid and Community Clouds), Service Platform as

a Service, Software as a Service with examples of services/service providers, models – Infrastructure as a Service, Cloud Reference model, Characteristics of Cloud Computing – a shift in paradigm Benefits and advantages of Cloud Computing. A brief introduction on Composability, Infrastructure, Platforms, Virtual Appliances,
Communication Protocols, Applications, Connecting to the Cloud by Clients.

Module 4: (6) IaaS – Basic concept, Workload, partitioning of virtual private server instances, Pods, aggregations, silos PaaS – Basic concept, tools and development environment with examples SaaS - Basic concept and characteristics, Open SaaS and SOA, examples of SaaS platform Identity as a Service (IDaaS) Compliance as a Service (CaaS). virtualization technologies : Types of virtualization (access, application, CPU, storage), Mobility patterns (P2V, V2V, V2P, P2P, D2C, C2C, C2D, D2D) Load Balancing and Virtualization: Basic Concepts, Network resources for load balancing, Advanced load balancing (including Application Delivery Controller and Application Delivery Network), Mention of The Google Cloud as an example of use of load balancing Hypervisors: Virtual machine technology and types, VMware vSphere Machine Imaging (including mention of Open Virtualization Format – OVF)

Text Book:

1. Internet of Things- Shriram K Vasudevan, Abhishek S Nagarajan, RMD Sundaram ISBN: 9788126578375, Wiley
2. Introduction to IoT by Sudip Mishra, Cambridge University Press.
3. Internet of Things and its Applications, Prof. Satish Jain (Author), Shashi Singh (Author), BPB Publication
4. Cloud Computing Bible by Barrie Sosinsky, Wiley India Pvt. Ltd, 2013
5. Mastering Cloud Computing by Rajkumar Buyya, Christian Vecchiola, S. Thamarai Selvi, McGraw Hill Education (India) Private Limited, 2013

Reference Books:

1. Internet of Things A to Z: Technologies and Applications, Qusay F. Hassan (Editor), Wiley 2. Internet of Things (IoT): Technologies, Applications, Challenges and Solutions, Tripathy, Anuradha, CRC Press
2. Embedded Systems and Robotics with Opensource tools, Dey, Mukherjee, CRC Press
3. Smart Computing with opensource platforms, Mukherjee, Dey, CRC Press
4. Cloud computing: A practical approach, Anthony T. Velte, Tata Mcgraw-Hill
5. Cloud Computing, Miller, Pearson
6. Building applications in cloud:Concept, Patterns and Projects, Moyer, Pearson
7. Cloud Computing – Second Edition by Dr. Kumar Saurabh, Wiley India

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Course Code : PEC-CS601D
Course Title : Parallel & Distributed Systems

Credit : 3

Course Outcomes

1. Remember the basics of formal and real distributed systems
2. Understand the the design issues and how to design a new distributed system
3. Apply knowledge to solve the design issues related to the distributed systems
4. Analyze specific distributed systems e.g. distributed database system (DDBS), Kerberos
5. Evaluate the specific challenges of DDBS while getting a formal notion of other types of distributed systems
6. Create new distributed systems and protocols

[Module 1] [6L]

1. Introduction to Distributed Systems in general [2L] [CO 1,2]
System Models, Nodes, Events, Messages, Channels, Local and Global States.
Algorithm writing techniques for distributed systems, Basics of complexity in distributed systems.
2. Clocks and Event Ordering [4L] [CO 2,3,5]
Global and Logical Clocks; Berkeley Algorithm, Network Time Protocol (NTP); Lamport's Clock, Vector Clock.

[Module 2] [11L]

3. Distributed Database Systems [8L] [CO 1,2,3,4,5]
Overview; Distributed Database Design,Semantics Data Control, Processing Issues;
4. Global Snapshots [3L] [CO 2,3,5]
Definition and Requirement; Chandy-Lamport's Algorithm

[Module 3] [13L]

5. Distributed Query Optimization, Transaction Management, Concurrency Control; [7L] [CO 2,3,4,5,6]
6. Leader Election [6L] [CO 2,3,5,6]
Definition and Requirement; Hirschberg-Sinclair Algorithm; Spanning Tree Construction; DFS tree construction

[Module 4] [12L]

7. Mutual Exclusion [6L] [CO 2,3,5]
Definition and Requirement; Permission based algorithm: Lamport's Algorithm, Ricart-Agrawala Algorithm, Maekawa's Algorithm; Token based algorithm:Suzuki Kasami Algorithm
8. Distributed File Systems [6L] [CO 2,3,4,5,6]
Definition and Requirement; DFS Architecture, Design Issues; Caching; Network File System (NFS)

[Advanced Topics] [6L]

1. Replication [2L] [CO 1,2,3]
Definition and Requirement; Design Issues in Replication; Consistency models: Sequential, Eventual, Casual;
2. Case study: Kerberos [2L] [CO 3,4,5]
3. Basics of Fault Tolerance [2L] [CO 1,2,3]
Definition and Requirement; Types of faults, Types of tolerance; Agreement Problems, Atomic Commits: 2-phase, 3-phase commits; Checkpointing and Recovery

[Books/ References]

1. Advanced Operating Systems by Mukesh Singhal and Nirajan Shivaratri
2. Distributed Systems: Principles and Paradigms by Andrew Tanenbaum and Maarten van Steen
3. Distributed Systems - An Algorithmic approach by Sukumar Ghosh
4. Design and Analysis of Distributed Algorithms by Nicola Santoro

5. Distributed Database Systems by D.Bell and J. Grimson, Addison

Course Code : PEC-CS601E
Course Title : Neural Network and Deep Learning
Credit : 3

Neural Network and Deep Learning
Credit - 3
Course Objective:
In this course, attendees will:

- Understand the context of neural networks and deep learning
- Know how to use a neural network
- Understand the data needs of deep learning
- Have a working knowledge of neural networks and deep learning
- Explore the parameters for neural networks

Prerequisites
Familiarity with Data Structure and Algorithm, Machine Learning,
Detailed Contents:
Module 1: Artificial Neural Network, activation function, multi-layer neural network, cardinality,
operations, and properties of fuzzy relations. Introduction History of Deep Learning,
Module 2: Feed forward neural network: Artificial Neural Network, activation function, multi-layer
neural network cardinality, operations, and properties of fuzzy relations. McCulloch Pitts Neuron-
Perceptron, Adaline model, Basic learning laws, Topology of neural network architecture.
Module 3: Training Neural Network: Risk minimization, loss function, back propagation, regularization, model selection, and optimization.
Module 4: Conditional Random Fields: Linear chain, partition function, Markov network, Belief
propagation, Training CRFs, Hidden Markov Model and Entropy.
Module 5: 5. Deep Learning: Deep Feed Forward network, regularizations, training deep models,
dropouts, Convolutional Neural Network, Recurrent Neural Network, Deep Belief Network.
Module 6: Deep Learning Applications Image Processing, Natural Language Processing, Speech
recognition, Video Analytics.
Text Books/Suggested References:
1. Goodfellow, I., Bengio,Y., and Courville, A., Deep Learning, MIT Press, 2016.
2. Yegnanarayana, B., Artificial Neural Networks PHI Learning Pvt. Ltd, 2009.
3. Golub, G.,H., and Van Loan,C.,F., Matrix Computations, JHU Press,2013.
4. Satish Kumar, Neural Networks: A Classroom Approach, Tata McGraw-Hill Education, 2004.
5. Ian Goodfellow, YoshuaBengio, Aaron Courville. Deep Learning, the MIT press, 2016
6. Bengio, Yoshua. “Learning deep architectures for AI." Foundations and trends in Machine Learning 2.1, Now Publishers, 2009
7. Deep Learning, Rajiv Chopra, Khanna Book Publishing, Delhi 2020.
8. Neural Network in computer Intelligence by Fu, TMH, 2003

CO1: Basic concepts of different aspects of Neural Network and Deep Learning
CO2: Understanding and interpreting various conceptual aspects of Neural Network and Deep Learning
CO3: Implementation of the concepts of Neural Network and Deep Learning
CO4: Analysing various concepts of Neural Network and Deep Learning
CO5: Evaluation of various concepts of Neural Network and Deep Learning

CO6: Generation models using various concepts of Neural Network and Deep Learning

Course Code : PEC-CS 602 D
Course Title : Data Mining and Analytics
Credit : 3

Course Outcomes	
Student will be able to	
CO1.	Study of different sequential pattern algorithms.
CO2.	Study the technique to extract patterns from time series data and it application in real world.
CO3.	Can extend the Graph mining algorithms to Web mining.
CO4.	Help in identifying the computing framework for Big Data

SYLLABUS:

Module I (Mapped to CO1)

Introduction to Data Warehousing; Data Mining: Mining frequent patterns, association and correlations; Sequential Pattern Mining concepts, primitives, scalable methods; Classification and prediction; Cluster Analysis – Types of Data in Cluster Analysis, Partitioning methods, Hierarchical Methods; Transactional Patterns and other temporal based frequent patterns,

Module II (Mapped to CO2)

Mining Time series Data, Periodicity Analysis for time related sequence data, Trend analysis, Similarity search in Time-series analysis;

Module III (Mapped to CO3)

Mining Data Streams, Methodologies for stream data processing and stream data systems, Frequent pattern mining in stream data, Sequential Pattern Mining in Data Streams, Classification of dynamic data streams, Class Imbalance Problem; Graph Mining; Social Network Analysis; modulation for communication, filtering, feedback control systems.
Web Mining, Mining the web page layout structure, mining web link structure, mining multimedia data on the web, Automatic classification of web documents and web usage mining; Distributed Data Mining.

Module IV (Mapped to CO4)

Introduction to Big Data Storage Platforms for Large Scale Data Storage, Introduction to Big Data Streaming Platforms for Fast Data, Introduction to Big Data Applications (Machine Learning), Recent trends in Distributed Warehousing and Data Mining, Class Imbalance Problem; Graph Mining; Social Network Analysis

Text Books & Reference Books:

1. Data Warehousing Fundamentals for IT Professionals, Second Edition by Paulraj Ponniah, Wiley India.
2. Data Warehousing, Data Mining, & OLAP – Second Edition by Alex Berson and Stephen J. Smith, Tata McGraw Hill Education

3. Data warehouse Toolkit by Ralph Kimball, Wiley India

4. Jiawei Han and M Kamber, Data Mining Concepts and Techniques,, Second Edition, Elsevier Publication, 2011.

5. Vipin Kumar, Introduction to Data Mining - Pang-Ning Tan, Michael Steinbach, Addison Wesley,2006.

6.. G Dong and J Pei, Sequence Data Mining, Springer, 2007.

Course Code : PEC-CS 602 E
Course Title : Image Processing and Pattern Recognition
Credit : 3

Course Outcomes Student will be able to	
CO1.	To know the fundamentals of digital image processing.
CO2.	To learn different image enhancement and restoration techniques.
CO3.	To learn image segmentation techniques and morphological processing.
CO4.	To learn feature extraction and different classification techniques.

SYLLABUS:

Module I :

Fundamentals of Digital Image Processing

Background, Digital Image Representation, Fundamental steps in Image Processing, Elements of Digital Image Processing - Image Acquisition, Storage, Processing, Communication, Display. A Simple Image Model, Geometric Model, Basic Transformation (Translation, Scaling, Rotation), Perspective Projection, Sampling & Quantization, Neighbour of pixels, Adjacency, Connectivity, Regions and boundaries, Distance Measures, Arithmetic/Logic Operations.

Module II :

Image Enhancement & Restoration

Spatial Domain Method, Frequency Domain Method, Contrast Enhancement -Linear & Nonlinear Stretching, Histogram Processing; Smoothing - Image Averaging, Mean Filter, Low-pass Filtering; Image Sharpening. Highpass Filtering, Highboost Filtering, Derivative Filtering, Homomorphic Filtering. Image Degradation/Restoration model, Noise models.

Module III :

Image Segmentation

Point Detection, Line Detection, Edge Detection, Combined Detection, Edge Linking & Boundary Detection– Local Processing, Global Processing via The Hough Transform; Thresholding-Foundation, Simple Global Thresholding, Optimal Thresholding; Region Oriented Segmentation - Basic Formulation, Region Growing by Pixel Aggregation, Region Splitting & Merging.

Morphological Image Processing

Erosion, Dilation, Duality, Opening & Closing, Basic Morphological Algorithms- Boundary Extraction, Hole Filling, Convex Hull, Thinning, Thickening, Skeletons.

Module IV :

Pattern Recognition

Feature extraction, Dimensionality reduction: Principal component analysis - it relationship to eigen analysis. Fisher discriminant analysis.

Supervised Classification – Nearest Neighbour, Support Vector Machine, Naïve Bayes, Discriminant Analysis, Decision Tree, Neural Network.

Unsupervised learning and clustering - Criterion functions for clustering. Algorithms for clustering: K-Means, Hierarchical, and other methods.

Text/Reference Books:

1. R C Gonzalez & R E Woods, Digital Image Processing, 3 rd /4 th Ed, PHI
2. A. K. Jain, Fundamentals of DIP, PHI
3. R C Ganzalez, R E Woods & S L Eddins, DIP using MATLAB, 2 nd Ed
4. R.O.Duda, P.E.Hart and D.G.Stork, Pattern Classification, John Wiley, 2001
5. S.Theodoridis and K.Koutroumbas, Pattern Recognition, 4th Ed., Academic Press, 2009
6. C.M.Bishop, Pattern Recognition and Machine Learning, Springer, 2006

Course Code : PEC-CS 602 F
Course Title : Social Media Analytics

Credit : 3

Course Outcome:

1. Students will be able to interpret the concept of social media analytics and its significance

2. Students will be able to apply a wide range of classification, clustering, estimation and prediction algorithms on Textual data.

3. Students will be able to perform social media network analysis to identify important social actors, subgroups and network properties in social media sites.

4. Students will be able to design new solutions to opinion extraction, sentiment classification and data summarization problems on realistic data sets.

SYLLABUS:

Module I :[10L]

Introduction to Social Media Analytics (SMA): Social media landscape, Need for SMA, Application of SMA in different areas. The social networks perspective - nodes, ties and influencers, Social network and web data and methods. Graphs and Matrices- Basic measures for individuals and networks, Information visualization.

Module II :[10L]

Mining Textual Data: Text Representation- tokenization, stemming, stop words, TF-IDF, Feature Vector Representation, NER,N-gram modeling. Text Clustering, Text Classification, Topic Modeling-LDA, HDP. Natural Language Processing Techniques for Micro-text Analysis.

Module III :[6L]

Social Media Network Analysis: Essentials of Social Graphs, Social Networks, Models, Information diffusion in social media. Web crawling and indexing, Link analysis, Random graphs and network evolution. Behavioral analytics, Influence and homophily, Recommendation in social media

Module IV :[10L]

Opinion Mining and Recent Trends: Sentiment classification, feature based opinion mining, comparative sentence and relational mining, Opinion spam. Recent trends in text, web and social media Analytics. Case studies on network analysis (Facebook, Twitter, Linkedin, Youtube, Google analytics).

Recommended Books:

1. Marshall Sponder, “Social Media Analytics" McGraw Hill, Latest Edition

2. Bing Liu, “Sentiment Analysis and Opinion Mining”, Morgan & Claypool Publishers, 2012.

3. Reza Zafarani, Mohammad Ali Abbasiand Huan Liu, “Social Media Mining-An Introduction”, Cambridge University Press, 2014.

4. Nitin Indurkhya, Fred J Damerau, “Hand book of Natural Language Process”, 2 nd Edition, CRC Press, 2010.

5. Bing Liu, “Web Data Mining – Exploring Hyperlinks, Contents, and Usage Data”, Springer, Second Edition, 2011.

6. Matthew A. Russell, “Mining the social web”, 2nd edition- O'Reilly Media, 2013.

Course Code : OEC-CS 601D
Course Title : Entrepreneurship and Start-up
Credit : 3

[COURSE OUTCOME]

1. Remembering the basic concept of Entrepreneurship.
2. Understanding different types and components of Entrepreneurship.
3. Apply that knowledge to understand Entrepreneurship and the Start-up market.
4. Analyse start-up ecosystems and strategies for implementing and it's uses in present market.
5. Evaluation of trends, opportunities and issues with Entrepreneurship.
6. Create new concepts to overcome Modern Entrepreneurship challenges.

Syllabus:

Module 1: Introduction to Entrepreneurship Meaning and concept of entrepreneurship, the history of entrepreneurship development, role of entrepreneurship in economic development, Myths about entrepreneurs, agencies in entrepreneurship management and future of entrepreneurship types of entrepreneurs [10L]

Module 2: The Entrepreneur: Why to become entrepreneur, the skills/ traits required to be an entrepreneur, Creative and Design Thinking, the entrepreneurial decision process, skill gap analysis, and role models, mentors and support system, entrepreneurial success stories. [8L]

Module 3: Communication: Importance of communication, barriers and gateways to communication, listening to people, the power of talk, personal selling, risk taking & resilience,
negotiation. [6L]

Module 4: Start-up Introduction to various form of business organization (sole proprietorship, partnership, corporations, Limited Liability company, mission, vision and strategy formulation.
[6L]

Module 5: Finance and Accounting concepts, Rate of Return Methods, Break even analysis, Financial statements, basic accounting concept. [10L]

Module 6: E-Cell: Meaning and concept of E-cells, advantages to join E-cell, significance of E-cell, various activities conducted by E-cell. [4L]

[Books/References]

1. The Age of Metapreneurship: A Journey into the Future of Entrepreneurship Design of Enterprise Systems: Theory, Architecture, and Methods, Ronald Giachetti.
2. The $100 Startup, Chris Guillebeau

3. Entrepreneurship: The Practice and Mindset Paperback, by Heidi M. Neck (Author), Christopher P. Neck (Author), Emma L. Murray.
4. ENTREPRENEURSHIP: The Art, Science, and Process for Success, Charles E. Bamford Associate Professor of Strategy & Entrepreneurship (Author), Garry D. Bruton Dr.

Course Code : OEC-CS 601E
Course Title : Financial and Cost Accounting
Credit : 3

SYLLABUS:

1. **Engineering Costs & Estimation** – Fixed, Variable, Marginal & Average Costs, Sunk Costs, Opportunity Costs, Recurring And Nonrecurring Costs, Incremental Costs, Cash Costs vs Book Costs, Life-Cycle Costs; Types Of Estimate, Estimating Models - PerUnit Model, Segmenting Model, Cost Indexes, Power-Sizing Model, Improvement & Learning Curve, Benefits. 6

2. **Cash Flow & Rate of Return Analysis** – Calculations, Treatment of Salvage Value, Annual Cash Flow Analysis, Analysis Periods; Internal Rate of Return, Calculating Rate of Return, Incremental Analysis; Best Alternative Choosing an Analysis Method, Future Worth Analysis, Benefit-Cost Ratio Analysis, Sensitivity and Breakeven Analysis. Economic Analysis In The Public Sector -Quantifying And Valuing Benefits & drawbacks. 6

3.**Present Worth Analysis:** End-Of-Year Convention, Viewpoint Of Economic Analysis Studies, Borrowed Money Viewpoint, Effect Of Inflation & Deflation, Taxes, Economic Criteria, Applying Present Worth Techniques, Multiple Alternatives. **Depreciation** - Basic Aspects, Deterioration & Obsolescence, Depreciation And Expenses, Types Of Property, Depreciation Calculation Fundamentals, Depreciation And Capital Allowance Methods, Straight-Line Depreciation Declining Balance Depreciation, Common Elements Of Tax Regulations For Depreciation And Capital Allowances. 6

4. **Replacement Analysis** - Replacement Analysis Decision Map, Minimum Cost Life of a New Asset, Marginal Cost, Minimum Cost Life Problems. **Accounting** – Function, Balance Sheet, Income Statement, Financial Ratios, Capital Transactions, Cost Accounting, Direct and Indirect Costs, Indirect Cost Allocation. 10

Course Code : HSMC 602
Course Title : ESP-VI
Credit Points : 2
Pre-Requisites : Basics of Data structure, Architecture, Theory of computation and Algorithms
L-T-P : 2-0-0 (Total Contact Hrs. 2)

Course Objective:

1. To learn about the Programming and Data Structures for professional exams
2. To learn about fundamentals of Computer Organization and Architecture for various exams
3. To learn about Theory of computation for professional exams
4. To learn about fundamentals of Algorithms and Database management systems for various exams

Course Outcomes At the end of the course the students will be able	
CO1.	To develop an understanding of Programming and Data Structures.
CO2.	To learn all types of Computer architecture and Organization fundamentals.
CO3.	To understand the theory of computation and its applications.
CO4.	To use fundamentals of Algorithms and Database principles.

Course Content:

Module No.	**Description**	**Hours**	**Blooms Level**	**PO(1..12) Mapping**
1.	**Programming and Data Structures:** Queues, linked lists, trees, binary search trees.	6	L1 (Remember) L2 (Understand) L4 (Analyse)	PO1,PO2, PO3
2.	**Computer Organization and Architecture:** Memory hierarchy: cache, main memory and secondary storage; I/O interface (interrupt and DMA mode)	12	L1 (Remember) L2 (Understand) L3 (Apply) L4 (Analyze)	PO1,PO2, PO3
3.	**Theory of Computation**: Regular expressions and finite automata. Context-free grammars and push-down automata. Regular and context free languages, pumping lemma. Turing machines and un-decidability	12	L1 (Remember) L2 (Understand) L3 (Apply) L4 (Analyze)	PO1,PO2, PO3
4.	**Algorithms and Database:** Searching, sorting, hashing. Asymptotic worst case time and space	6	L1 (Remember) L2	PO1,PO2, PO3

	complexity. Algorithm design techniques: greedy, dynamic programming and divide-and-conquer. Graph search, minimum spanning trees, shortest paths. ER-model. Relational model: relational algebra, tuple calculus, SQL.		(Understand) L3 (Apply) L4 (Analyze)	

Learning Resources:

Reference Books:

1. G.K publishers GATE Computer Science Engineering,
2. McGraw hill GATE 2020 Computer Science Engineering,
3. Wiley GATE 2020 Computer Science Engineering,

Course Code : PCC-CS691
Course Title : Compiler Design Lab
Credit Points : 2

Prerequisites: Basic Computation & Principles of Computer Programming, Data structures and algorithms, Formal Language & Automata Theory, Basics of C/C++

COURSE OUTCOME:

CO1: Understand the practical approaches of how a compiler works.

CO2: Understand and analyze the role of syntax and semantics of Programming languages in compiler construction.

CO3: Apply the techniques and algorithms used in Compiler Construction in compiler component design.

CO4: To use different tools in construction of the phases of a compiler for the mini language.

1. Familiarization with LEX by writing simple specifications for tokens such as, identifiers, numbers, comments in C/C++, etc. All LEX specifications must be compiled and executed with appropriate inputs. (**2L**)

2. LEX specification for tokens of the small language in ALSU‟s book (**1L**)

3. Complete the specifications in (2) above to make a complete lexical analyzer. (**1L**)

4. Familiarization with YACC by writing simple specifications for desk calculator, variable declarations in C (only numbers and array). All YACC specifications must be compiled and executed with appropriate inputs. Note that this exercise also requires LEX specifications o the tokens involved. (**2L**)

5. YACC specifications for the syntax of the small language in ALSU‟s book (**1L**)

6. Adding error recovery to (5) above to make a complete parser. (**1L**)

7. S-attributed specification of the semantics of the small language in ALSU‟s book to be incorporated into YACC specifications produced in (6) above. (**3L**)

8. Adding semantic error recovery to the semantic analyzer in (7) above to make a complete semantic analyzer. (**1L**)

9. Intermediate code generation for the constructs of the small language in ALSU‟s book to be incorporated into the semantic analyzer of (8) above. (**3L**)

Text Books and/or Reference Material

Text Books

1. Compilers: Principles, Techniques, and Tools , by A.V. Aho, Monica Lam, Ravi Sethi, and J.D. Ullman, (2nd ed.), Addison-Wesley, 2007 (main text book, referred to as ALSU in lab assignments).
2. K.D. Cooper, and Linda Torczon, Engineering a Compiler, Morgan Kaufmann, 2004.

Reference Books

1. K.C. Louden, Compiler Construction: Principles and Practice, Cengage Learning, 1997.
2. D. Brown, J. Levine, and T. Mason, LEX and YACC, O‟Reilly Media, 1992.

Course Code : PCC-CS692
Course Title : Computer Networks Lab
Credit Points : 2

Prerequisites: Programming Languages

Module 1 [6L]

1. Getting started with Basics of Network configurations files and Networking Commands in Linux.
2. To familiarize and understand the use and functioning of System Calls used for Operating system and network programming in Linux.
3. Familiarization and implementation of programs related to Process and thread. 4. Implement the First Readers-Writers Problem.

Module 2 [6L]

4. Implement programs for Inter Process Communication using PIPE, Message Queue and Shared Memory.
5. Implement Client-Server communication using Socket Programming and TCP and UDP as transport layer PROTOCOL. *
6. Implement a multi user chat server using TCP as transport layer protocol. *
7. Implement Concurrent Time Server application using UDP to execute the program at

remote server. Client sends a time request to the server; server sends its system time back to the client. Client displays the result. *

Module 3 [6L]

Implement and simulate algorithm for Distance vector routing protocol. Implement and simulate algorithm for Link state routing protocol.
Using Wireshark observe data transferred in client server communication using UDP and identify the UDP datagram.
Using Wireshark observe Three Way Handshaking Connection Establishment, Data Transfer and Three-Way Handshaking Connection Termination in client server communication using TCP.

Module 4 [6L]

Develop a packet capturing and filtering application using raw sockets.
Design and configure a network with multiple subnets with wired and wireless LANs using required network devices. Configure the following services in the network- TELNET, SSH, FTP server, Web server, File server, DHCP server and DNS server.*
Install network simulator GNS3 in any of the Linux operating system and simulate wired and wireless scenarios.

Text Books:

1. Data Communication and Networking, 4th Edition, Behrouz A. Forouzan, McGrawHill.
2. Data and Computer Communication, 8th Edition, William Stallings, Pearson Prentice Hall India.
3. Routing and Switching, Cisco Press

Reference Books:

1. Computer Networks, 8th Edition, Andrew S. Tanenbaum, Pearson New International Edition.
2. Internetworking with TCP/IP, Volume 1, 6th Edition Douglas Comer, Prentice Hall of India.
3. TCP/IP Illustrated, Volume 1, W. Richard Stevens, Addison-Wesley, United States of America.

Course Code : PCC-CS693
Course Title : Cloud Computing and IoT Lab
Credit Points : 1
Pre-requisites:
Students should have prior knowledge of Computer Networks and Programming for Problem Solving (C Language)

Course Outcome:

CO 1: Students can able to understand the various concepts of the IoT and their technologies.
CO 2: Students can able to develop the IoT application using different hardware platforms.

CO 3: Students can able to implement the various IoT Protocols and deploy the IoT application into a cloud environment.
CO 4: Students can able to understand the basic principles of cloud computing.

Syllabus:

Module 1:
Assignment 1. Creating a EC2 Instance in AWS.
Assignment 2. Deployment of a web application in AWS.
Assignment 3. Deployment of an Analytics system in AWS-EC2.

Module 2:
Assignment 4. Study and Install IDE of Arduino and different types of Arduinos, program using Arduino IDE for Blink LED, Sensor Interfacing, Actuator Interfacing
Assignment 5. WLAN communication with Node-MCU, Bluetooth and Zigbee and Raspberry Pi.

Module 3:
Assignment 6. Study and implement MQTT protocol using Arduino/Python (MQTT-Paho).
Assignment 7. Study and Configure Raspberry Pi and Node-Red.
Assignment 8. Raspberry Pi and Node-Red based MQTT Application development.

Module 4:
Assignment 9. Study and Implement Device to cloud application using Raspberry Pi, IBM Watson/Bluemix IoT platform
Assignment 10. Fundamental application of RTSP, RTMP streaming protocols.

Text Books and/or Reference Books:

1. Cloud Computing Bible by Barrie Sosinsky, Wiley India Pvt. Ltd, 2013
2. Mastering Cloud Computing by Rajkumar Buyya, Christian Vecchiola, S. Thamarai Selvi, McGraw Hill Education (India) Private Limited, 2013
3. Cloud computing: A practical approach, Anthony T. Velte, Tata Mcgraw-Hill 4. Cloud Computing, Miller, Pearson
5. Building applications in cloud: Concept, Patterns and Projects, Moyer, Pearson 6. Cloud Computing – Second Edition by Dr. Kumar Saurabh, Wiley India
7. "The Internet of Things: Enabling Technologies, Platforms, and Use Cases", by Pethuru Raj and Anupama C. Raman, CRC Press.
8. Adrian McEwen, Designing the Internet of Things, Wiley, 2013.

Course Code	**:**	HSMC682
Course Title	**:**	**SKILL DEVELOPMENT FOR PROFESSIONALS - VI**
Credit Points	**:**	1
Pre-Requisites	**:**	Basic Mathematics, General English from primary to high school.
L-T-P	:	2-0-0 (Total Contact Hrs. 2)

Course Objective:

1. To enhance the aptitude & analytical skill of students with multiple tricky approaches.
2. To prepare the students for various competitive examinations & professional exams

Course Outcomes	
CO1.	The ability to communicate effectively with a range of audiences.
CO2.	The ability to face the test and interview conducted by different companies and succeed. And also preparation to appear different competitive exams starts.
CO3.	The ability to recognize the need for continuing professional development.
CO4.	The ability to succeed in competitive exams (BANK/IBPS/SSC/GATE / GRE / PSU's/Placement Aptitude etc.).

SYLLABUS:

Module No.	Description	Hours
1.	**Revision and Advanced Problems in Quantitative Aptitude:** 1)Numbers (+, -, x, etc), Percentages, Ratio, Partnership, Linear Equations, Profit & Loss 2)Averages, Mixtures & Allegations, Number System, Time and Work 3)Simple & Compound Interest, Other / Misc Quantitative Apt., Indices and Surds, Quadratic Equations 4)Permutations & Combinations, Probability, Geometry, Mensuration 5)Data Interpretation, Various Charts, Diagrams, Tables	12
2.	**Revision and Advanced Problems in Reasoning** 1)Coding, Series & Numbers, Blood Relations, Analogy 2)Cubes, Data Sufficiency, Non-Verbal Reasoning 3)Syllogisms, Puzzles, Machine I/O, Inequality 4)Seating Arrangement, Calendar / Clock 5)Statements, Other / Misc Logical Reasoning, Decision Making (Ethics)	12
3.	**Revision and Advanced Questions in Verbal English** 1)Grammar, 2)Clauses, 3)Spotting errors, 4)Sentence Correction, 5)Blanks, 6)Reading Comprehensions, 7)Vocabulary	12

Learning Resources:

Reference Books:

1. Objective General English- S.P Bakshi
2. English Grammar and Competition-S.C Gupta
3. Fast Track Objective Arithmetic- Rajesh Verma
4. Advance Maths- Rakesh Yadav
5. Verbal and Non-Verbal Reasoning- R.S Agarwal
6. A new approach to Reasoning- BS Sijwali
7. Quantitative Aptitude-R.S Agarwal

Semester VII (Fourth year] Curriculum								
Sl. No.	**Type of course**	**Code**	**Course Title**	**Hours per week**				**Credits**
				Lecture	**Tutorial**	**Practical**	**Sessional**	
Theory Papers								
1	Professional Core courses	PCC-CS701	Network Security & Cryptography	2	0	0	0	2
2	Professional Elective courses	PEC-CS701	Professional Elective-IV E: Human Computer Interaction F: Cyber Security & Blockchain	3	0	0	0	3
3	Professional Elective courses	PEC-CS702	Professional Elective-V E: Big Data Analytics F: Speech and Natural Language Processing	3	0	0	0	3
4	Open Elective courses	OEC – CS701	Open Elective-II A: Soft Skill & Interpersonal Communication F: Comparative Study of Literature	3	0	0	0	3
5	Humanities and social sciences including Management	HSMC702	Essential Studies for Professionals – VII	2	0	0	0	2
	Total			**13**	**0**	**0**	**0**	**13**
Practical Papers								
1	Innovative Project	PROJ-CS701	Project-II	**0**	**0**	**12**	**0**	**6**
	Total			**0**	**0**	**16**		7
Sessional Papers								
1	Humanities and social sciences including Management	HSMC782	Skill Development for Professionals - VII	0	0	0	2	1
2	Mandatory Additional Requirements (MAR)	MAR781	Mandatory Additional Requirements (MAR) -VII	0	0	0	0	0
3	MOOCs (Mandatory for Honours)	MOOCs 721	Massive Open Online Course 7.1(Mandatory for B.Tech (Honours))	0	0	0	3	3
4		MOOCs 722	Massive Open Online Course 7.2(Mandatory for B.Tech (Honours))	0	0	0	2	2
	Total			**0**	**0**	**0**	**7**	**6**
	Total			**13**	**0**	**12**	**7**	**26**

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Course Code : PCCCS701
Course Title : **Network Security & Cryptography**
Credit Points : 2

Course Outcome:

At the end of the course the students will be able

1. To understand Symmetrical and Asymmetrical cryptography along with Data integrity, Authentication, Digital Signatures. Describe network security services and mechanisms.
2. To identify security at different network layers.
3. To understand firwalls and intrusion detection and prevention systems.
4. To learn about different types of Malicious Software and Software Security.

Module1 [12L]

Introduction to Cryptographic Tools: Security Goals, Threats, Vulnerabilities and Attacks, Types of Attacks, Security Services and Mechanisms, Symmetric Key Cryptography, Block Ciphers, Stream Ciphers, DES, IDEA, AES, Asymmetric Key Cryptography, RSA, ElGamal Cryptosystems, Elliptic Curve Cryptosystems, Message Integrity and Message Authentication, Cryptographic Hash Functions, Digital Signatures, Key Management, Kerberos.

Module2 [12L]

Security at different Network Layers : IP Security Overview, IP Security Architecture, Security Association, Authentication Header, Encapsulated Security Payload, Internet Key Exchange, SSL Architecture, Four SSL Protocols, SSL Message Format, Transport Layer Security, HTTPS, SSH, Email Security, PGP, S/MIME.

Module3 [8 L]

Firewalls and Intrusion Detection and Prevention Systems: Firewall Characteristics, Types of Firewalls, Firewall Basing, Firewall Location and Configuration, Intrusion Detection Systems: Intruders, Intrusion Detection, Host Based Intrusion Detection, Network Based Intrusion Detection, Distributed Intrusion Detection, Intrusion Detection Exchange Format, Honeypots, Intrusion Prevention Systems.

Module4 [8L]

Malicious Software and Software Security: Types of Malicious Software, Viruses, Virus Countermeasures, Worms, Bots, Botnets, Rootkits, Buffer Overflow Attacks, Defence against Buffer Overflows, Handling Program Inputs, Writing Safe Program Codes, Interaction with Operating System and Other Software.

Text Books and/or Reference Material

1. Computer Security: Principles and Practices, by William Stallings and Larry Brown, First Edition, 2008, Pearson Education

2. Network Security: Private Communication in a Public World by Charlie Kaufman, Radia Perlman and Mike Speciner, Second Edition, 2003, Prentice Hall India 21
3. Cryptography and Network Security, by William Stallings, Fifth Edition,Prentice Hall, 2010
4. Cryptography and Network Security by Behrouz A. Forouzan and Debdeep Mukhopadhyay, Second Edition, 2010, Tata McGraw Hill
5. Network Security Essentials: Applications and Standards, by William Stallings, Edition, , Pearson Education
6. Security in Computing, by Charles P. Pfleeger, Shari Lawrence Pfleeger, 4th Edition, 2007, Prentice Hall

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Human Computer Interaction
Code: PEC-CS701 E
Contacts: 3L

Course Outcome:

CO1: Remembering the principles of designing user interfaces

CO2: Understanding the concepts of interface and visual design

CO3: Apply the concepts of usability, user experience and user-centered design

CO4: Analyze the interactivity of designed interface

CO5: Evaluate the cognitive aspects of multimedia presentation

CO6: Create user interface for specific project

Introduction [Human factors, Fundamentals of Human perception, Human skill level and Behaviour, Dialogues and tasks, Framework for HCI, Modeling Human Computer Interaction];

Human Computer Interface Design [Information Design, Interaction Design and Sensorial design,Guidelines for user interface design, Dialogue Design, Graphic Design and Style issues, Standard interface elements in windows, Interface Design tool: MATLAB];

Structured System Analysis and Design Methodology (SSADM) [Methodology for Dialog design,Prototyping, Prototyping tools];

Visual Design [Introduction, Visual Rhetoric, Organising information, Factors designers consider when creating illustration and visual design, Designing for screen, Typography for computer screen, Spatial relationships in the interface, Symbols & Semiotics in the interface. Visual design methodology : (Clarity, Consistency, Appearance), Visual Coding, Layout Principles];

Cognitive aspect in Multimedia Presentation [Cognitive domain of learning, Knowledge and Skill, Retention, Learning Style, Affective and Cognitive domain learning, Role of the creator of Multimedia learning material, Presentation format, Interactivity, System Quality, Media mix, Cognitive issues in user interface]

Cyber Security and Blockchain
Code: PEC-CS701 F
Contacts: 3L

Introduction to Cybercrime and Laws:

Definition and Origins of Cybercrime, information Security, Who are Cybercriminals? Classifications of Cybercrimes. How Criminals Plan Them – Introduction, How Criminals Plan the Attacks, Cybercafe and Cybercrimes, Botnets, Attack Vector, The Indian IT ACT 2000 and amendments.

Tools and Methods used in Cybercrime Introduction, Proxy Server and Anonymizers, Password Cracking, Keyloggers and Spyware, Virus and Warms, Trojan and backdoors, Steganography, DOS and DDOS attack, SQLinjection, Buffer Overflow.

2 Phishing and Identity Theft: Introduction, Phishing – Methods of Phishing, Phishing Techniques, Phishing Toolkits and Spy Phishing. Identity Theft – PII, Types of Identity Theft, Techniques of ID Theft. Digital Forensics Science, Need for Computer Cyber forensics and Digital Evidence, Digital Forensics Life Cycle.

3 Introduction: Overview of Block chain, Public Ledgers, Bitcoin, Smart Contracts, Block in a Block chain, Transactions, Distributed Consensus, Public vs. Private Block chain, Understanding Crypto currency to Block chain, Permissioned Model of Block chain, Overview of Security aspects of Block chain, Basic Crypto Primitives: Cryptographic Hash Function, Properties of a hash function, Hash pointer and Merkle tree, Digital Signature, Public Key Cryptography, A basic cryptocurrency.

4 Bitcoin and Block chain: Creation of coins, Payments and double spending, Bitcoin Scripts, Bitcoin P2P Network, Transaction in Bitcoin Network, Block Mining, Block propagation and block relay. Working with Consensus in Bitcoin: Distributed consensus in open environments, Consensus in a Bitcoin network, Proof of Work (PoW) – basic introduction, Hashcash PoW, Bitcoin PoW, Attacks on PoW and the monopoly problem, Proof of Stake, Proof of Burn and Proof of Elapsed Time, The life of a Bitcoin Miner, Mining Difficulty, Mining Pool.

Big Data Analytics
PECCS702E
Contacts: 3L

Course Outcomes: After completion of course, students would be able to:

CO1: Basic concepts of different aspects of Big Data Analytics
CO2: Understanding and interpreting various conceptual aspects of Big Data Analytics
CO3: Implementation of the concepts of Big Data Analytics
CO4: Analysing various concepts of Big Data Analytics
CO5: Evaluation of various concepts of Big Data Analytics
CO6: Generation models using various concepts of Big Data Analytics

Module 1: Introduction to big data

Introduction to BigData Platform, Traits of Big data, Challenges of Conventional Systems, Web Data, Evolution of Analytic Scalability, Analysis vs Reporting, Statistical Concepts: Sampling Distributions, Re-Sampling, Statistical Inference, Prediction Error.

Module 2: Basic data analysis and data analytic methods

Regression Modelling, Multivariate Analysis, Bayesian Modelling, Inference and Bayesian Networks, Support Vector and Kernel Methods, Analysis of Time Series: Linear Systems Analysis, Nonlinear Dynamics, Rule Induction, Neural Networks: Learning and Generalization, Competitive Learning, Principal Component Analysis and Neural Networks, Fuzzy Logic: Extracting Fuzzy Models from Data Fuzzy Decision Trees, Stochastic Search Methods.

Module 3: Frequent item sets and clustering

Mining Frequent item sets: Market Based Model, Apriori Algorithm, Handling Large Data Sets in Main Memory, Limited Pass Algorithm, Counting Frequent item sets in a Stream, Clustering Techniques: Hierarchical, K-Means, Frequent Pattern based Clustering Methods.

Module 4: Mining data streams

Introduction to Streams Concepts: Stream Data Model and Architecture, Stream Computing, Sampling Data in a Stream: Filtering Streams, Counting Distinct Elements in a Stream, Estimating Moments, Counting Oneness in a Window, Decaying Window, Real time Analytics Platform (RTAP) Applications, Case Studies, Real Time Sentiment Analysis, Stock Market Predictions.

Module 5: Framework, technologies, tools and visualization

Map Reduce: Hadoop, Hive, MapR, Sharding, NoSQL Databases: S3, Hadoop Distributed File Systems, Visualizations: Visual Data Analysis Techniques, Interaction Techniques; Systems and Analytics Applications, Analytics using Statistical packages, Industry challenges and application of Analytics

Text Books/Suggested References:

1. Bart Baesens, "Analytics in a Big Data World: The Essential Guide to data Science and its Applications", Wiley publications, 2014.
2. V.K. Jain, Big Data & Hadoop, Khanna Book Publishing Co., Delhi. (ISBN 978-93-82609-131)
3. Michael Berthold, David J. Hand, "Intelligent Data Analysis", Springer, 2003.
4. Anand Rajaraman and Jeffrey David Ullman, "Mining of Massive Datasets", Cambridge University Press, 2020.
5. Jeeva Jose, Beginner's Guide for Data Analysis using R Programming, Khanna Book Publishing House, 2019.
6. Bill Franks, "Taming the Big Data Tidal Wave: Finding Opportunities in Huge Data Streams with Advanced Analytics", Wiley, 2012.

Speech and Natural Language Processing

PEC-CS 702 F

Contacts: 3L

Course Outcomes

After successfully completing the course the student should be able to

1. Remember different aspects of speech and natural language processing systems

2. Understand the techniques to process the Human Languages Such as English and other Indian Languages.

3. Apply different language techniques to achieve the required goals analyse, evaluate, create

4. Develop and analyse computational methods for real world applications like machine translation.

5. Evaluate which computer programs will be suitable for a given language technological problem, install them and apply them to linguistic data.

6. Create Automatic Speech Recognition system with new features to improve the performance of the existing systems.

Introduction:

Introduction to various levels of natural language processing, Ambiguities and computational challenges in processing various natural languages. Introduction to Real life applications of speech and natural language processing such as spell and grammar checkers, information extraction, question answering, and machine translation, automatic speech recognition, dialogue processing etc.

Text Processing:

Character Encoding, Word Segmentation, Sentence Segmentation, Introduction to Corpora, Corpora Analysis.

Lexical Syntax:

Introduction to word types, POS Tagging, Maximum Entropy Models for POS tagging, Multi-word Expressions.

Language Modelling:

The role of language models. Simple N-gram models. Estimating parameters and smoothing. Evaluating language models.Syntax and Semantics:

Introduction to phrases, clauses and sentence structure, Shallow Parsing and Chunking, Shallow Parsing with Conditional Random Fields (CRF), Lexical Semantics, Word Sense Disambiguation, WordNet, Thematic Roles, Semantic Role Labelling with CRFsApplication of NLP

NL Interfaces, Text Summarization, Sentiment Analysis, Machine Translation, Question answering.Phonetics: Speech Sounds and Phonetic Transcription, Articulatory Phonetics, Phonetic Resources.

Automatic Speech Recognition:

The Automatic Speech Recognition Task, Feature Extraction for ASR, Speech Recognition Architecture

ASR Evaluation: Word Error Rate

Text Books

1. Daniel Jurafsky and James H. Martin “Speech and Language Processing”, 3rd edition, Prentice Hall, 2009.

Reference Books

1. Chris Manning and HinrichSchütze, "Foundations of Statistical Natural Language Processing", 2nd edition, MITPress Cambridge, MA, 2003.

2. NitinIndurkhya, Fred J. Damerau "Handbook of Natural Language Processing", Second Edition, CRC Press, 2010. 3. James Allen "Natural Language Understanding", Pearson Publication 8th Edition. 2012

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Soft Skills & Interpersonal Communication

Code: OEC-IT701A

Contact: 3L

Course Outcomes:

CO1: Remember the basic concepts of Soft Skills, including self discovery and motivation, various norms of interpersonal communication, interpersonal skills including interview skills.
CO2: Understand concepts to all-round personalities with a mature outlook to function effectively in different circumstances.
CO3: Apply the concepts of soft skills to develop broad career plans, evaluate the employment market, identify the organizations to get good placement, match the job requirements and skill sets.

CO4: Analyse effectively in various selection procedures adopted by the recruiters.

CO5: Evaluate different strategies adopted for effective communication.

CO6: Create effective technical correspondence and prepare presentation which will produce results.

Module 1

1. Soft Skills: An Introduction – Definition and Significance of Soft Skills; Process, Importance and Measurement of Soft Skill Development.

2. Self-Discovery: Discovering the Self; Setting Goals; Beliefs, Values, Attitude, Virtue.

3. Positivity and Motivation: Developing Positive Thinking and Attitude; Driving out Negativity; Meaning and Theories of Motivation; Enhancing Motivation Levels.

Module 2

1. Interpersonal Communication: Interpersonal relations; communication models, process and barriers; team communication; developing interpersonal relationships through effective communication; listening skills; essential formal writing skills; corporate communication styles – assertion, persuasion, negotiation.

2. Public Speaking: Skills, Methods, Strategies and Essential tips for effective public speaking.

Module 3

1. Group Discussion: Importance, Planning, Elements, Skills assessed; Effectively disagreeing, Initiating, Summarizing and Attaining the Objective.

2. Non-Verbal Communication: Importance and Elements; Body Language.

3. Teamwork and Leadership Skills: Concept of Teams; Building effective teams; Concept of Leadership and honing Leadership skills.

Module 4

1. Interview Skills: Interviewer and Interviewee – in-depth perspectives. Before, During and After the Interview. Tips for Success.

2. Presentation Skills: Types, Content, Audience Analysis, Essential Tips – Before, During and After, Overcoming Nervousness.

3. Etiquette and Manners – Social and Business.

4. Time Management – Concept, Essentials, Tips.

5. Personality Development – Meaning, Nature, Features, Stages, Models; Learning Skills; Adaptability Skills.

Text and References:

1. Communication Skills and Soft Skills (English, Mixed media product, Kumar E. Suresh)

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Subject Code : HSMC 702	**Category**:
Subject Name : ESP-VII	**Semester** : 7th
L-T-P : 2-0-0 (Total Contact Hrs. 2)	**Credit:** 2
Pre-Requisites: Basics of Compiler, Operating systems and Computer network	

Course Objective:

1. To learn about the Details of compiler designs for professional exams
2. To learn about fundamentals of data base management for various exams
3. To learn about Operating systems for professional exams
4. To learn about fundamentals computer networks for various exams

Course Outcomes:

At the end of the course the students will be able

1. To develop a detailed knowledge of compiler designs.
2. To learn all types of Data Base Management Systems' fundamentals.
3. To understand Operating Systems and its applications.
4. To use fundamentals of Computer networks and its methods.

Course Content:

Module No.	Description	Hours
1.	**Compiler Design** Lexical analysis, parsing, syntax-directed translation. Runtime environments. Intermediate code generation. Local optimization, Data flow analyses: constant propagation, likeness analysis, common sub expression elimination	6
2.	**Databases** Integrity constraints, normal forms. File organization, indexing (e.g., B and B+ trees). Transactions and concurrency control	12
3.	**Operating System:** Processes, threads, inter-process communication, concurrency and synchronization, Deadlock, CPU scheduling. Memory management and virtual memory, File system. Flow and error control techniques, switching. IPv4/IPv6, routers and routing algorithms (distance vector, link state). TCP/UDP and sockets, congestion control. Application layer protocols (DNS, SMTP, POP, FTP, HTTP). Basics of Wi-Fi.	12
4.	**Computer Networks:** Concept of layering: OSI and TCP/IP Protocol Stacks; Basics of packet, circuit and virtual circuit- switching; Data link layer: framing, error detection, Medium Access Control, Ethernet bridging; Routing protocols: shortest path, flooding, distance vector and link state routing; Fragmentation and IP addressing, IPv4, CIDR notation, Basics of IP support protocols (ARP, DHCP, ICMP), Network Address Translation (NAT); Transport layer: flow control and congestion control, UDP, TCP, sockets; Application layer protocols: DNS, SMTP, HTTP, FTP, Email.	6

Learning Resources:

Reference Books:

1. G.K publishers GATE Computer Science Engineering,
2. McGraw hill GATE 2020 Computer Science Engineering,
3. Wiley GATE 2020 Computer Science Engineering,

Subject Code : HSMC782	**Category**:
Subject Name : **SKILL DEVELOPMENT FOR PROFESSIONALS - VII**	**Semester** : 7TH
L-T-P : 2-0-0 (Total Contact Hrs. 2)	**Credit:** 1
Pre-Requisites: Basic Mathematics, General English from primary to high school.	

Course Objective:

1. To enhance the aptitude & analytical skill of students with multiple tricky approaches.
2. To prepare the students for various competitive examinations & professional exams

Course Outcomes:

1. Revision of entire Syllabus along with specific type of Mock Tests for the competitive

exams like UPSC, IBPS, SBI PO/SO, SSC,RAIL, INSURANCE etc.

2. Learning Advance quantitative Aptitude Techniques on Algebra , Geometry, Menstruation, Tr etc which is very essential for appearing in different competitive Exams along with SSC.
3. Solving Puzzle based problems & learning different Analytical techniques.
4. Misc Practise sets on different Competitive Exams mains level papers.

Course Content:

Module No.	Description	Hours
1.	**Quantitative Aptitude** Miscellaneous Problems on quantitative aptitude [CAT level-4].	12
2.	**Logical Reasoning:** Miscellaneous Problems on Logical Reasoning [CAT level 4].	12
3.	**Verbal English** Miscellaneous Problems on Verbal English [CAT level-4].	12

Learning Resources:

Reference Books:

1. Objective General English- S.P Bakshi
2. English Grammar and Competition-S.C Gupta
3. Fast Track Objective Arithmetic- Rajesh Verma
4. Advance Maths- Rakesh Yadav
5. Verbal and Non-Verbal Reasoning- R.S Agarwal
6. A new approach to Reasoning- BS Sijwali
7. Quantitative Aptitude-R.S Agarwal

1.

Semester VIII (Fourth year] Curriculum								
Sl. No.	**Type of course**	**Code**	**Course Title**	**Hours per week**				**Credits**
				Lecture	**Tutorial**	**Practical**	**Sessional**	
Theory Papers								
1	Basic Science Course	BSC801	Biology	3	0	0	0	3
2	Professional Elective courses	PEC-CS801	Professional Elective-V E: Information Retrival F: Data Science	3	0	0	0	3
3	Open Elective courses	OEC - CS801	Open Elective-III E: Human Resource Development & Organizational Behaviour F: Enterprise System	3	0	0	0	3
4	Open Elective courses	OEC - CS802	Open Elective-IV D: Mobile Computing E: Real Time Operating System	3	0	0	0	3
5	Humanities and social sciences including Management	HSMC802	Essential Studies for Professionals - VIII	2	0	0	0	2
	Total			**11**	**0**	**0**	**0**	**11**
Practical Papers								
1	Innovative Project	PROJ-CS801	Project-III	0	0	12	0	6
	Total			**0**	**0**	**12**	**0**	**6**
Autonomous Papers								

1	Humanities and social sciences including Management	HSMC882	Skill Development for Professionals - VIII	0	0	0	2	1
2	Mandatory Additional Requirements (MAR)	MAR881	Mandatory Additional Requirements (MAR)-VIII	0	0	0	0	0
3	MOOCs (Mandatory for Honours)	MOOCs 821	Massive Open Online Course 8.1 (Mandatory for B.Tech (Honours))	0	0	0	3	3
4		MOOCs 822	Massive Open Online Course 8.2 (Mandatory for B.Tech (Honours))	0	0	0	2	3
5	Grand viva	PCC-CS881	Grand Viva-voce	0	0	0	0	2
	Total			**11**	**0**	**0**	**7**	**9**
	Total			**11**	**0**	**12**	**7**	**26**

Paper Name: Biology

Paper Code: BSC801

Credit: 3

Lecture Hours:

2-1-0 Syllabus:

Module 1. (2 hours)- Introduction

Purpose: To convey that Biology is as important a scientific discipline as Mathematics, Physics and Chemistry. Bring out the fundamental differences between science and engineering by drawing a comparison between eye and camera, Bird flying and aircraft. Mention the most exciting aspect of biology as an independent scientific discipline. Why we need to study biology? Discuss how biological observations of 18th Century that lead to major discoveries. Examples from Brownian motion and the origin of thermodynamics by referring to the original observation of Robert Brown and Julius Mayor. These examples will highlight the fundamental importance of observations in any scientific inquiry.

Module 2. (3 hours)- Classification

Purpose: To convey that classification per se is not what biology is all about. The underlying criterion, such as morphological, biochemical or ecological be highlighted. Hierarchy of life forms at phenomenological level. A common thread weaves this hierarchy Classification. Discuss classification based on (a) cellularity- Unicellular or multicellular (b) ultrastructure-

prokaryotes or eucaryotes. (c) energy and Carbon utilization -Autotrophs, heterotrophs, lithotropes (d) Ammonia excretion – aminotelic, uricoteliec, ureotelic (e) Habitata-acquatic or terrestrial (e) Molecular taxonomy- three major kingdoms of life. A given organism can come under different category based on classification. Model organisms for the study of biology come

from different groups. E.coli, S.cerevisiae, D. Melanogaster, C. elegance, A. Thaliana, M. musculus

Module 3. (4 hours)-Genetics

Purpose: To convey that ―Genetics is to biology what Newton‘s laws are to Physical Sciences‖ Mendel‘s laws, Concept of segregation and independent assortment. Concept of allele. Gene mapping, Gene interaction, Epistasis. Meiosis and Mitosis be taught as a part of genetics. Emphasis to be give not to the mechanics of cell division nor the phases but how genetic material passes from parent to offspring. Concepts of recessiveness and dominance. Concept of mapping of phenotype to genes. Discuss about the single gene disorders in humans. Discuss the concept of complementation using human genetics.

Module 4. (4 hours)-Biomolecules

Purpose: To convey that all forms of life has the same building blocks and yet the manifestations are as diverse as one can imagine Molecules of life. In this context discuss monomeric units and polymeric structures. Discuss about sugars, starch and cellulose. Amino acids and proteins. Nucleotides and DNA/RNA. Two carbon units and lipids.

Module 5. (4 Hours). Enzymes

Purpose: To convey that without catalysis life would not have existed on earth Enzymology: How to monitor enzyme catalyzed reactions. How does an enzyme catalyzereactions. Enzyme classification. Mechanism of enzyme action. Discuss at least two examples. Enzyme kinetics and kinetic parameters. Why should we know these parameters to understand biology? RNA catalysis.

Module 6. (4 hours)- Information Transfer

Purpose: The molecular basis of coding and decoding genetic information is universal Molecular basis of information transfer. DNA as a genetic material. Hierarchy of DNA structure- from single stranded to double helix to nucleosomes. Concept of genetic code. Universality and degeneracy of genetic code. Define gene in terms of complementation and recombination.

Module 7. (5 hours). Macromolecular analysis
Purpose: How to analyses biological processes at the reductionistic level Proteins- structure and function. Hierarch in protein structure. Primary secondary, tertiary and quaternary structure. Proteins as enzymes, transporters, receptors and structural elements.

Module 8. (4 hours)- Metabolism

Purpose: The fundamental principles of energy transactions are the same in physical and biological world. Thermodynamics as applied to biological systems. Exothermic and endothermic versus endergonic and exergoinc reactions. Concept of K and its relation to standard

free energy. Spontaneity. ATP as an energy currency. This should include the breakdown of glucose to COeq2 + H O (Glycolysis and Krebs cycle) and synthesis of glucose from CO2 and H O (Photosynthesis). Energy yielding and energy consuming reactions. Concept of Energy charge.

Module 9. (3 hours)- Microbiology

Concept of single celled organisms. Concept of species and strains. Identification and classification of microorganisms. Microscopy. Ecological aspects of single celled organisms. Sterilization and media compositions. Growth kinetics.

References:

1) Biology: A global approach: Campbell, N. A.; Reece, J. B.; Urry, Lisa; Cain, M, L.; Wasserman, S. A.; Minorsky, P. V.; Jackson, R. B. Pearson Education Ltd
2) Outlines of Biochemistry, Conn, E.E; Stumpf, P.K; Bruening, G; Doi, R.H., John Wiley and Sons
3) Principles of Biochemistry (V Edition), By Nelson, D. L.; and Cox, M. M.W.H. Freeman and Company
4) Molecular Genetics (Second edition), Stent, G. S.; and Calender, R.W.H. Freeman and company,
Distributed by Satish Kumar Jain for CBS Publisher
5) Microbiology, Prescott, L.M J.P. Harley and C.A. Klein 1995. 2nd edition Wm, C. Brown Publishers

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Paper Name: Information Retrieval
Code: PEC-CS801D
Credit: 3
Professional Elective I

Course Outcome:

1. Students will be able to remember the concepts of Information Retrieval architecture with the essence of different models.

2. Students will be able to understand the concepts of Tokenizing, Indexing.

3. Students will be able to apply the different concepts of Web Search and Categorization with different experimental evaluations.

4. Students will be able to analyse the process of Clustering and Language Model based translation techniques.

5. Students will be able to evaluate different recommender systems as well as different information extraction processes and question answering systems.

6. Students will be able to create one intelligent and personalized IR system with different Machine and Deep Learning techniques.

Detailed Syllabus: Module No.	Content	Hrs./Module
1	**Introduction:** Goals and history of IR. The impact of the web on IR. **Basic IR Models**: Boolean and vector-space retrieval models; ranked retrieval; text-similarity metrics; TF-IDF (term frequency/inverse document frequency) weighting; cosine similarity.	5
2	**Basic Tokenizing, Indexing, and Implementation of Vector-Space Retrieval**: Simple tokenizing, stop-word removal, and stemming; inverted indices; efficient processing with sparse vectors; Java implementation. **Experimental Evaluation of IR**: Performance metrics: recall, precision, F-measure, and NDCG; Evaluations on benchmark text collections.	6
3	**Query Operations**: Relevance feedback; Query expansion. **Text Representation**: Word statistics; Zipf's law; Porter stemmer; morphology; index term selection; using thesauri. **Web Search**: Search engines; spidering; meta crawlers; directed spidering; link analysis (e.g. hubs and authorities, Google PageRank); shopping agents.	8
4	**Text Categorization**: Categorization algorithms: Rocchio, nearest neighbour, and naive Bayes. Applications to information filtering and organization. **Text Clustering**: Clustering algorithms: agglomerative clustering; k-means; expectation maximization (EM).	8

Text Books and Reference Books:

1. Christopher D. Manning, Prabhakar Raghavan and Hinrich Schütze, Introduction to Information Retrieval, Cambridge University Press. 2008.

2. Cheng Xiang Zhai, Statistical Language Models for Information Retrieval (Synthesis Lectures Series on Human Language Technologies), Morgan & Claypool Publishers, 2008.

Paper Name: Data Science

Code: PEC-CS801E

Credit: 3

PRE-REQUISITES

Introduction to Programming

Probability

OBJECTIVES

The objective of this course is to impart necessary knowledge of the mathematical foundations needed for data science and develop programming skills required to build data science applications.

LEARNING OUTCOMES

At end of this course, the students will be able to:

Demonstrate understanding of the mathematical foundations needed for data science.

Collect, explore, clean, munge and manipulate data.

Implement models such as k-nearest Neighbors, Naive Bayes, linear and logistic regression, decision trees, neural networks and clustering.

Build data science applications using Python based toolkits.

DETAIL CONTENTS

1. Introduction to Data Science (4 Hours)

Concept of Data Science, Traits of Big data, Web Scraping, Analysis vs Reporting

2. Introduction to Programming Tools for Data Science (6 Hours)

2.1 Toolkits using Python: Matplotlib, NumPy, Scikit-learn, NLTK

2.2 Visualizing Data: Bar Charts, Line Charts, Scatterplots

2.3 Working with data: Reading Files, Scraping the Web, Using APIs (Example: Using

the Twitter APIs), Cleaning and Munging, Manipulating Data, Rescaling,

Dimensionality Reduction 3. Mathematical Foundations (12 Hours)

3.1 Linear Algebra: Vectors, Matrices,

3.2 Statistics: Describing a Single Set of Data, Correlation, Simpson's Paradox,

Correlation and Causation

3.3 Probability: Dependence and Independence, Conditional Probability, Bayes's

Theorem, Random Variables, Continuous Distributions, The Normal Distribution,

The Central Limit Theorem

3.4 Hypothesis and Inference: Statistical Hypothesis Testing, Confidence Intervals, Phacking, Bayesian Inference

4. Machine Learning (16 Hours)

Overview of Machine learning concepts – Over fitting and train/test splits, Types of Machine learning – Supervised, Unsupervised, Reinforced learning, Introduction to Bayes Theorem, Linear Regression- model assumptions, regularization (lasso, ridge, elastic net), Classification and Regression algorithms- Naïve Bayes, K-Nearest Neighbors, logistic regression, support vector machines (SVM), decision trees, and random forest, Classification Errors, Analysis of Time Series- Linear Systems Analysis, Nonlinear Dynamics, Rule Induction, Neural NetworksLearning And Generalization,

Overview of Deep Learning.

5. Case Studies of Data Science Application (6 Hours)

Weather forecasting, Stock market prediction, Object recognition, Real Time Sentiment Analysis.

LIST OF PRACTICALS

1. Write a programme in Python to predict the class of the flower based on available attributes.

2. Write a programme in Python to predict if a loan will get approved or not.

3. Write a programme in Python to predict the traffic on a new mode of transport.

4. Write a programme in Python to predict the class of user.

5. Write a programme in Python to indentify the tweets which are hate tweets and which are not.

6. Write a programme in Python to predict the age of the actors.

7. Mini project to predict the time taken to solve a problem given the current status of the user

LIST OF SUGGESTED BOOKS

1. Joel Grus, "Data Science from Scratch: First Principles with Python", O'Reilly Media

2. Aurélien Géron, "Hands-On Machine Learning with Scikit-Learn and Tensor Flow: Concepts, Tools, and Techniques to Build Intelligent Systems", 1st Edition, O'Reilly Media

3. Jain V.K., “Data Sciences”, Khanna Publishing House, Delhi.

4. Jain V.K., “Big Data and Hadoop”, Khanna Publishing House, Delhi.

5. Jeeva Jose, “Machine Learning”, Khanna Publishing House, Delhi.

6. Chopra Rajiv, “Machine Learning”, Khanna Publishing House, Delhi.

7. Ian Goodfellow, Yoshua Bengio and Aaron Courville, "Deep Learning", MIT Press

http://www.deeplearningbook.org

8. Jiawei Han and Jian Pei, "Data Mining Concepts and Techniques", Third Edition, Morgan

Kaufmann Publishers

Paper Name: Human Resource Development and Organizational Development
Code: OEC-CS801D
Credit: 3
Course:

Module 1-Human Resource Management: Meaning & Definition, Functions, Scope & Objectives, Qualities of a HR Manager

Module 2- Human Resource Planning: Meaning & Definition, Importance of HRP, HRP Process. Barriers of HRP, Factors of sound HRP. Recruitment – Meaning & Definition, Sources of Recruitment, Recruitment Process, Effective Recruitment. Training & Performance Appraisal- Definition & Objective, Areas of Training, Meaning & Definition of Performance Appraisal, process, Effective principles of performance Appraisal.

Module 3- Industrial Relations: Concept & Meaning, Objective & Importance, Reasons of poor Industrial Relation. Industrial Disputes- Meaning & Definition, Causes of Industrial Dispute, Prevention of Industrial Dispute, Conditions for good Industrial Relation.

Module 4- Workers Participation in Management: Meaning & Need, Forms of Participation, Scheme of participation, Merits & Demerits. Collective Bargain- Meaning & Definition, Objective & Importance, Process of Collective Bargain, Effective Condition. Employee Discipline-Guidelines for action, Penalties & Punishment, Rewards of Discipline.

Text Books:

1. Human Resource Management. P. Subba Rao, Himalaya Publishing House, 2012.

2. Human Resource Management. K.Aswathappa. Mc GRAW HILL Education, 2013.

Reference Books:

1. Human Resource Development Management. A. M.Seikh S.Chand, 2003.

2. Human Resource Management. S.S.Khanka, S. Chand, 2014.

Course Code : OEC-CS 801E
Course Title : Enterprise System

Credit : 3

[Course Outcomes]

1. Remembering the basic concept of enterprise systems.
2. Understanding different types and components of enterprise systems.
3. Apply that knowledge to understand architectures and the enterprise systems market.
4. Analyse system integration and strategies for implementing and using enterprise systems.
5. Evaluation of trends, opportunities and issues with enterprise systems.
6. Create new concepts to overcome Modern Enterprise system challenges.

Module 1: Fundamentals of enterprise systems, introduces concepts: Enterprise Resource Planning, Supply Chain Management, Customer Relationship Management, Business Process Management. Organisational behaviour. [10L]

Module 2: Evolution of enterprise systems and provide a theoretical understanding of enterprise systems in organisation, Enterprise resource planning and utilisation.
[8L]

Module 3: Enterprise Systems implementation and use, enterprise systems implementation life cycle and use in organisations. Case-studies, insight opportunities and challenges of enterprise systems in organisations. hands-on experiences with enterprise systems applications, Communication: Use, advantages and barriers.
[8L]

Module 4: Contemporary issues in Enterprise Systems practice, analyse an enterprise systems implementation, Implementation of new tools and technology to overcome challenges, solutions or change measures. [8L]

Module 5: Finance and Accounting concepts, Rate of Return Methods, Break even analysis, Financial statements, basic accounting concept. [10L]

[Books/References]

1. Enterprise Systems for Management, J. Thompson.
2. Design of Enterprise Systems: Theory, Architecture, and Methods, Ronald Giachetti.
3. Enterprise Supply Chain Management, V Sehgal
4. Enterprise Performance Management Done Right: An Operating System for Your Organization, Ron Dimon
5. The Flexible Enterprise (Flexible Systems Management), by Sushil and Edward A. Stohr.

Course Code : OEC-CS 802D

Course Title : Mobile Computing

Credit : 3

Course Objective:

In this course, attendees will:

• Understand the basic concepts of mobile computing

• Be familiar with the network protocol stack

• Learn the basics of mobile telecommunication system

• Be exposed to Ad-Hoc networks

• Gain knowledge about different mobile platforms and application development

Prerequisites

Computer Networks, DBMS

Course Outcome:

CO1: Basic concepts of different aspects of Mobile Computing

CO2: Understanding and interpreting various conceptual aspects of Mobile Computing

CO3: Implementation of the concepts of Mobile Computing

CO4: Analysing various concepts of Mobile Computing

CO5: Evaluation of various concepts of Mobile Computing

CO6: Generation models using various concepts of Mobile Computing

Detailed Contents:

UNIT I: INTRODUCTION (9)

Mobile Computing – Mobile Computing Vs wireless Networking – Mobile Computing Applications – Characteristics of Mobile computing – Structure of Mobile Computing Application. MAC Protocols – Wireless MAC Issues – Fixed Assignment Schemes – Random Assignment Schemes – Reservation Based Schemes.

UNIT II MOBILE INTERNET PROTOCOL AND TRANSPORT LAYER (9) Overview of Mobile IP – Features of Mobile IP – Key Mechanism in Mobile IP – route Optimization. Overview of TCP/IP – Architecture of TCP/IP- Adaptation of TCP Window – Improvement in TCP Performance.

UNIT III MOBILE TELECOMMUNICATION SYSTEM (9) Global System for Mobile Communication (GSM) – General Packet Radio Service (GPRS) – Universal Mobile Telecommunication System (UMTS).

UNIT IV MOBILE AD-HOC NETWORKS (9)

Ad-Hoc Basic Concepts – Characteristics – Applications – Design Issues – Routing – Essential of Traditional Routing Protocols –Popular Routing Protocols – Vehicular Ad Hoc networks (VANET) – MANET Vs VANET – Security.

UNIT V MOBILE PLATFORMS AND APPLICATIONS (9)

Mobile Device Operating Systems – Special Constrains & Requirements – Commercial mobile Operating Systems – Software Development Kit: iOS, Android, BlackBerry, Windows Phone – MCommerce – Structure – Pros & Cons – Mobile Payment System –Security-issues.

Text/Reference Book:

1. "Pervasive Computing", Burkhardt, Pearson
2. "Mobile Communication", J. Schiller, Pearson
3. "Wireless and Mobile Networks Architectures", Yi-Bing Lin & Imrich Chlamtac, John Wiley & Sons, 2001
4. "Mobile and Personal Communication systems and services", Raj Pandya, Prentice Hall of India.
5. "Guide to Designing and Implementing wireless LANs", Mark Ciampa, Thomson learning, Vikas Publishing House, 2001.
6. "Wireless Web Development", Ray Rischpater, Springer Publishing,
7. "The Wireless Application Protocol", Sandeep Singhal, Pearson.
8. "Third Generation Mobile Telecommunication systems", by P.Stavronlakis, Springer Publishers

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Course Code : OEC-CS 802E
Course Title : Real Time Operating System

Credit : 3

Course Outcome

1. Students will be able to remember the concepts of OS, BIOS, BOOT, Threads, Scheduling, Process etc.
2. Students will be able to understand the different RTOS concepts.
3. Students will be able to apply the different RTOS process management concepts.
4. Students will be able to analyse the different IPC concepts.
5. Students will be able to evaluate different memory management aspects.
6. Students will be able to create different segments of RTOS by knowing the different real life case studies.

Detailed Syllabus:

Module No.	Content	Hrs./Module

1	**INTRODUCTION:** Introduction to Operating System: Computer Hardware Organization, BIOS and Boot Process, Multi-threading concepts, Processes, Threads, Scheduling.	10
2	**BASICS OF REAL-TIME CONCEPTS:** Terminology: RTOS concepts and definitions, real-time design issues, examples, Hardware Considerations: logic states, CPU, memory, I/O, Architectures, RTOS building blocks, Real-Time Kernel.	13
3	**PROCESS MANAGEMENT:** Concepts, scheduling, IPC, RPC, CPU Scheduling, scheduling criteria, scheduling algorithms Threads: Multi-threading models, threading issues, thread libraries, synchronization Mutex: creating, deleting, prioritizing mutex, mutex internals	13
4	**INTER-PROCESS COMMUNICATION:** Messages, Buffers, mailboxes, queues, semaphores, deadlock, priority inversion,	6
5	**PIPES MEMORY MANAGEMENT:** Process stack management, run-time buffer size, swapping, overlays, block/page management, replacement algorithms, real-time garbage collection	6
6	**CASE STUDIES:** Case study Linux POSIX system, RTLinux / RTAI, Windows system, Vxworks, uItron Kernel Design Issues: structure, process states, data structures, inter-task communication mechanism, Linux Scheduling	12

Text Books and Reference Books:

1. J. J Labrosse, “MicroC/OS-II: The Real –Time Kernel”, Newnes, 2002.

2. Jane W. S. Liu, “Real-time systems”, Prentice Hall, 2000.

3. W. Richard Stevens, “Advanced Programming in the UNIX® Environment”, 2nd Edition, Pearson Education India, 2011.

Subject Code : HSMC 802	**Category**:
Subject Name : ESP-VIII	**Semester** : 8th
L-T-P : 2-0-0 (Total Contact Hrs. 2)	**Credit:** 2
Pre-Requisites: Basic Social Science from primary to high school, NCERTs	

Course Objective:

1. To learn about basic of History to know about our past and to implement it in our daily life.
2. To learn about the Political System of Our Country.
3. To learn the concepts of Basics of Geography and Economics from which Students will acquire knowledge for Competitive exams.

Course Outcomes:

At the end of the course the students will be able

1. To inculcate human values and ethical thinking among students.
2. To prepare the stage for facing different levels of civil service and other competitive examinations.
3. To prepare the ground for making them aware of the happenings, cultural historical and developmental aspects of the country as well as global affairs

4. Learning current affairs with technique.

Course Content:

Module No.	Description	Hours
1.	**Laws of Society**: Evolution of Indian Constitution, Part -II and Part - III.	6
2.	**Our Freedom Struggle**: Indian National Congress, National Movement- 1905- 1947.	12
3.	**Know Your Country**: Physiography of India.	12
4.	**Economics, India and World and Universal Human Values:** Capital and Money Market, Fiscal System of India. Monthly Current Affairs Magazine, Nature acceptance of Human Values, Competence in Professional Ethics, Strategies for transition towards value based life and profession.	6

Learning Resources:

Text Books:

1. NCERT Books from class 8-12.

Reference Books:

1. Indian Constitution- M.Laxmikant
2. Indian Economy-Ramesh Singh
3. History of Modern India- Bepan Chandra
4. Geography of India- Majid Hussain
5. Current Affairs Magazine of IEM-UEM

Subject Code : HSMC882		**Category**:	
Subject Name : **SKILL DEVELOPMENT FOR PROFESSIONALS - VIII**		**Semester**	: 8TH
L-T-P	: 2-0-0 (Total Contact Hrs. 2)	**Credit:** 1	
Pre-Requisites: Basic Mathematics, General English from primary to high school.			

Course Objective:

1. To enhance the aptitude & analytical skill of students with multiple tricky approaches.
2. To prepare the students for various competitive examinations & professional exams

Course Outcomes:

1. To make the students aware of all the nuances of various public sector examinations.
2. To motivate them hone their previously learnt skills necessary for cracking various exams like civil service examination (UPSC and State PSC), Staff Selection Commission, Railway Services and other exams.
3. This part of the syllabus will also expertise them to boost their conversational skills by allowing them to speak on a variety of topics with ease.
4. Appearing mock tests of different competitive exams both prelims & mains.

Course Content:

Module No.	Description	Hours
1.	**Mock tests OF UPSC CSAT-II.**	12

Learning Resources:

Reference Books:

1. Verbal and Non-Verbal Reasoning- R.S Agarwal
2. Quantitative Aptitude-R.S. Agarwal
3. GK PUBLICATION –for UPSC Civil Services